Abelian Varieties

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Introduction

Let E be an elliptic curve over k (i.e. a proper irreducible curve over k, whose points form a group). We quote the following facts from the elliptic curve course:

1. $E \hookrightarrow \mathbb{P}^2_k$ is a nonsingular cubic.
2. Group law is commutative
3. Suppose k is algebraically closed, $E(k)$ is a divisble group. If $n \geq 1$, $E[n] = \{x \in E(k) : nx = 0\}$ the n-th torsion points are finite. Moreover: if $\mathrm{char}(k) = 0$ or $\mathrm{char}(k) \nmid n \implies E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$; if $\mathrm{char}(k) = p$, $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$ or 0.
4. $E(k) \cong \mathrm{Cl}^0(E)$, $p \mapsto (p) - (0)$. This is Abel's theorem.

Roughly speaking, abelian varieties are higher dimensional generalizations of elliptic curves: proper variety X/k, whose points form a group. There are analogues of (1)-(4):

1. X is projective
2. Group law is commutative
3. if $k = \overline{k}$,$X(k)$ is a divisble group. $X[n] \cong (\mathbb{Z}/n\mathbb{Z})^{2\cdot\dim(X)}$, if $\mathrm{char} \nmid n$; if $p = \mathrm{char}(k) > 0$, $X[p^r] = (\mathbb{Z}/p^r\mathbb{Z})^j$, for some $0 \leq j \leq \dim(X)$.
4. There exists certain "dual variety" $\hat{X}$, such that $\hat{X} \cong \mathrm{Pic}^0(X) \subseteq \mathrm{Pic}(X)$. There is a surjective homomorphism $X \to \hat{X}$ with finite kernel.

1 Varieties Over a Field

Definition. Let k be field, and fix an algebraic closure $\overline{k}$. A *k-variety* is a k-scheme X such that:

1. X is separated and finite type
2. X is geometrically integral, i.e $X \times_{\mathrm{Spec}\ (k)} \mathrm{Spec}\ (\overline{k})$ is integral.

Example. *Let $k = \mathbb{R}, X = \mathrm{Spec}\ (\mathbb{R}[T_1, T_2]/({T_1}^2 + {T_2}^2))$ is not geometrically irreducible. Let $k = \mathbb{F}_p(t), X = \mathrm{Spec}\ k[T_1, T_2]/({T_1}^p - t{T_2}^p)$ is not geometrically reduced.*

Proposition 1.1. *If X, Y are k-varieties, then so is their product.*

Proof. This essentially follows from the fact that tensor product of domains is a domain. □

Suppose X is a k-variety, with generic point $\eta \in X$. Then we define the function field of X to be $K(X) = \mathcal{O}_{X,\eta}$. This is a finitely generated k-algebra, since X is of finite type.

Proposition 1.2. *Let X be a k-variety, then k is algebraically closed in $K(X)$.*

Proof. Suppose not, choose $k \subsetneq k(a) \subseteq K(X)$ for a algebraic over k. Since $K(X) = \mathcal{O}_{X,\eta}$, there exists an open affine $\emptyset \neq \text{Spec } (A) = U \subseteq X$ such that $a \in A \subseteq K(X)$. Now $X_{\overline{k}} \supseteq \text{Spec } (A \otimes \overline{k})$, and $A \otimes \overline{k} \supseteq k(a) \otimes \overline{k}$ is not a domain. So $A \otimes \overline{k}$ is not a domain, contradiciton. □

Remark. *There exists X integral of finite type over k, separated, but not geometrically integral such that k is algebraically closed in $K(X)$. Note this cannot happen if k is perfect.*

We can prove that $\mathbb{P}^n_{\mathbb{Z}} \to \text{Spec } (\mathbb{Z})$ is proper, and base change proper maps are proper. So the projective spaces are proper over k, and so are the projective k-varieties.

Theorem 1.3. *Suppose X is proper scheme over* Spec (A) *with A noetherian. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathcal{F})$ is a finite A-module.*

Corollary 1.4. *Suppose X is proper variety, then $\Gamma(X, \mathcal{O}_X) = k$*

Proof. We know $\Gamma(X, \mathcal{O}_X) = B$ is a finite dimensional k-algebra, hence $B \subseteq K(X)$ as X integarl. So B is a finite field extension inside $K(X)$, this means $B = k$. □

There are a lot of proper varieties not projective.

Definition. A morphism $f : X \to Y$ of integral schemes is birational if there exists a nonempty open V in Y such that: $f^{-1}(V) \to V$ is an isomorphism.

Theorem 1.5 (Chow's lemma)**.** *If X is proper k-variety, then there exists a birational $f : X' \to X$ for X' projective.*

Proof. Let $X = \cup_{i=1}^m U_i$ be an open affine cover. Consider $U_i \hookrightarrow \mathbb{A}^{n_i}_k \hookrightarrow \mathbb{P}^{n_i}_k$. Let $U_i^* \subseteq \mathbb{A}^{n_i}_k$ be the closure of U_i, with the reduced subscheme structure. Then $U_i^* \cap \mathbb{A}^{n_i}_k = U_i$ and U_i^* is a variety.

Now let $U = \cap_{i=1}^m U_i \hookrightarrow \prod \mathbb{A}^{n_i}_k$, this is a closed immersion: $U_1 \cap U_2 = U_1 \times_X U_2 \subseteq U_1 \times_k U_2$ is closed. Let U^* be the closure of U in $\mathbb{P} = \prod \mathbb{P}^{n_i}_k$, we then have the commutative diagram:

$$\begin{array}{ccccc} U & \hookrightarrow & U^* & \hookrightarrow & \mathbb{P} \\ \downarrow & & \downarrow{\scriptstyle P_i} & & \downarrow \\ U_i & \hookrightarrow & U_i^* & \hookrightarrow & \mathbb{P}^{n_i}_k \end{array}$$

Let $V_i^* = P_i^{-1}(U_i) \subseteq U^*, U^* = \cup V_i^*$ and $V_i^* \cap V_j^* \to U_i \cap U_j$ can be given by either p_i or p_j. So p_i glue to get $U^* \xrightarrow{f} X$, which is identity on U. Set $X' = U^*$. □

Remark. *There's a general version of Chow's lemma. Let X be any noetherian scheme, $g : X \to S$ proper morphism. Then there exists a surjective S-morphism $f : X' \to X$ such that X' is a closed subscheme of $\mathbb{P}^n_S$, and there exists an open dense $U \subseteq X$ such that $f : f^{-1}(U) \to U$ is an isomorphism.*

Theorem 1.6 (Nagata's theorem)**.** *Every k-variety is an open subscheme of a proper k-variety.*

2 Differentials, Nonsingularity, Smoothness

Definition. Let $\varphi : A \to B$ be a ring homomorphism. We have $\Omega_{B/A}$ a B-module of differentials, and a universal A-derivation $d_{B/A} : B \to \Omega_{B/A}$ such that:

$$\mathrm{Hom}_B(\Omega_{B/A}, M) \simeq \mathrm{Der}_A(B, M)$$

Here are two constructions of the differential.

1. $\Omega_{B/A} = P/Q$, where P is the free B-module on symbols $\{[b] : b \in B\}$; Q is the submodule generated by $[\varphi(a)], a \in A$ and $[b_1 b_2] - b_1[b_2] - b_2[b_1], b_i \in B$. We define $d_{B/A}(b) = [b](\bmod Q)$.

2. $B \otimes_A B \xrightarrow{\mu} B$, with kernel I, and thus I/I^2 is a $B \otimes B$-module. Now there two ways to restrict the scalar $B \to B \otimes B, b \mapsto b \otimes 1$ or $1 \otimes b$, but the define the same B-module structure on $\Omega_{B/A} \coloneqq I/I^2$, since $b \otimes 1 - 1 \otimes b \in I$. Moreover, the differential is given by $d' : B \to \Omega_{B/A}, b \mapsto 1 \otimes b - b \otimes 1$.

Proposition 2.1. $P/Q \simeq I/I^2, [b](\bmod Q) \mapsto d'(b)$.

Remark. *Let $C = B \otimes_A B/I^2$ be a B-algebra. Note $\mu : C \to B$ has a splitting $b \mapsto b \otimes 1$. This gives a B-algebra isomorphism $B \oplus I/I^2 \cong C$, where the algebra structure on LHS is given by $(b, f) * (b', f') = (bb', bf' + b'f)$.*

The construction of differential is functorial:

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

there is an induced map of B-morphism between differentials: $\Omega_{B/A} \to \Omega_{B'/A'}$, transitive for $B \to B' \to B''$. Note this moreover induces $\Omega_{B/A} \otimes_B B' \to \Omega_{B'/A'}$.

Proposition 2.2. *If $B' = B \otimes_A A'$, then: $\Omega_{B/A} \otimes_A A' \simeq \Omega_{B/A} \otimes_B B' \simeq \Omega_{B'/A'}$. In particular, suppose $S \subset A$ multiplicative, $B' = B \otimes_A S^{-1}A = S_B^{-1}B$ is the localization of B at image of S, we then have: $\Omega_{S_B^{-1}B/S^{-1}A} = S_B^{-1} \cdot \Omega_{B/A}$.*

Example. *Let $B = A[t_1, \cdots, t_n]$, then $\Omega_{B/A} = \oplus B dt_i$ is free.*

Proposition 2.3. *Let $A \to B \to C$ be ring homomorphisms. There is an exact sequence of C-modules: $\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$, where the middle map is given by $d_{C/A}(c) \mapsto d_{C/B}(c)$*

Proposition 2.4. *Let $A \to B \to C = B/I$ are ring homomorphisms. There is an exact sequence of C-modules:*

$$I/I^2 \to \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0$$

where the first map is given by $[f] \mapsto df \otimes 1$.

Corollary 2.5. *Suppose $B = A[x_i : i \in I], C = B/(\{f_i\})$. Then $\Omega_{C/A} = \bigoplus C \cdot dx_i / \sum C \cdot df_j$.*

Example. *Let L/K be finit field extension. Then $\Omega_{L/K} = 0 \iff L/K$ is separable.*

Let $f : X \to Y$ be a morphism of schemes. We wish to define a quasi-coherent sheaf $\Omega_{X/Y}$ of $\mathcal{O}_X$-module, and a $f^{-1}\mathcal{O}_Y$-linear map $d = d_{X/Y} : \mathcal{O}_X \to \Omega_{X/Y}$. We wish to have the following properties:

1. Functoriality.

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

   This induces $g'^*(\Omega_{X/Y}) \to \Omega_{X'/Y'}$. This is an isomorphism if g, g' are open immersions.

2. if $X = \text{Spec } B, Y = \text{Spec } A$, $\Omega_{X/Y} = \widetilde{\Omega_{B/A}}$

Recall that if $X = \text{Spec } R$ and M is a R-module, we can associate a corresponding $\tilde{M}$ of $\mathcal{O}_X$-module with global section M. Now if $f : \text{Spec } R' \to \text{Spec } R$, $f^*\widetilde{M} = \widetilde{M \otimes_R R'}$, in particular has global section $M \otimes_R R'$.

We now start the construction of the differentials. First we assume f is separated.

The diagonal map $\Delta : X \to X \times_Y X \coloneqq X^2$ is a closed immersion. This induces a quasicoherent sheaf of ideals $\mathcal{I}_\Delta \subseteq \mathcal{O}_{X^2}$, with $\mathcal{O}_{X^2}/\mathcal{I}_\Delta = \Delta_*\mathcal{O}_X$. Then $\mathcal{I}_\Delta/\mathcal{I}_\Delta^2$ is a quasi-coherent module for $\Delta_*\mathcal{O}_X$. Therefore, there exists a unique $\mathcal{O}_X$-module $\Omega_{X/Y}$ such that $\mathcal{I}_\Delta/\mathcal{I}_\Delta^2 = \Delta_*\Omega_{X/Y}$.

Now if $j : X' \hookrightarrow X, Y' \subseteq Y, f(X') \subseteq Y'$, we want to show that $\Omega_{X'/Y'} = \Omega_{X/Y}|_{X'} = j^*\Omega_{X/Y}$. In fact, if $\Delta' : X' \hookrightarrow X' \times_{Y'} X' = X' \times_Y X'$, then $\mathcal{I}_{\Delta'} = \mathcal{I}_\Delta|_{X' \times X'} \implies \mathcal{I}_{\Delta'}/\mathcal{I}_{\Delta'}^2 = \mathcal{I}_\Delta/\mathcal{I}_\Delta^2|_{X' \times X'}$, this checks the first property. The second property is clear from construction.

For general f, the diagonal map is not necessarily a closed immersion, but it is still an immersion (closed immersion followed by open immersion): $X \hookrightarrow V \hookrightarrow X \times_Y X$ for V closed subscheme of $X \times_Y X$. Now $\mathcal{I}_{X \subset V}$ is a quasicoherent $\mathcal{O}_V$-module. As before we define $\Omega_{X/Y} = i^*(\mathcal{I}_{X \subset V}/\mathcal{I}_{X \subset V}^2)$. One can show this is independent of the choice of V.

It remains to define $d : \mathcal{O}_X \to \Omega_{X/Y}$, a $f^{-1}\mathcal{O}_Y$-linear morphism. It suffices to define on each open affine patch $U = \text{Spec } B \subseteq X$ such that $f(U) \subseteq \text{Spec } A \subseteq Y$, which is equivalent to give $\mathcal{O}_X(U) = B \to \Omega_{X/Y}(U) = \Omega_{B/A}$. But we can just take $d_{B/A}$ as before.

Remark. *Not every immersion can be written as an open immersion followed by a closed immersion, though the counterexample is hard to construct.*

We can rewrite the two exact sequences in terms of sheaves:

1. Suppose we are given: $X \xrightarrow{f} Y \to S$. then we have an exact sequence:
$$f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$$
2. Suppose we have closed immersion: $i : Z \hookrightarrow X$ given by a quasicoherent sheaf of ideals $\mathcal{I} = \mathcal{I}_{Z/X} \subseteq \mathcal{O}_X$, and $f : X \to Y$ morphism, we have an exact sequence:
$$\mathcal{I}/\mathcal{I}^2 \to i^*\Omega_{X/Y} \to \Omega_{Z/Y} \to 0$$

Finiteness: recall if X is locally Noetherian, then a quasi-coherent $\mathcal{O}_X$-module is coherent if for any open affine Spec $(R) = U \subseteq X$, $\mathcal{F}_U = \tilde{M}$ for $M = \mathcal{F}(U)$ a finite R-module. Again it suffices to check for elements of a single open affine cover. The definition of coherent sheaf for X not locally noetherian is a little bit more complicated.

Proposition 2.6. *Let $f : X \to Y$ locally of finite type, Y locally Noetherian. Then $\Omega_{X/Y}$ is coherent.*

Remark. *Hypothesis and Hilbert basis theorem implies that X is locally Noetherian.*

Proof. This statement is local on X. So we can assume $X =$ Spec $B, Y =$ Spec A and A noetherian, $B = A[T_1, \cdots, T_n]/I$. It suffices to prove that $\Omega_{B/A}$ is a finite B-module. By the second exact sequence, there exists a surjection:
$$\Omega_{A[T_1,\cdots,T_n]/A} \otimes B = \bigoplus BdT_i \twoheadrightarrow \Omega_{B/A}$$
□

We will now define the notion of tangent and cotangent spaces. Let X be a k-scheme, $x \in X$ any point with residue field $k(x)$ a field extension of k. We also write x for the corresponding morphism of k-scheme $x :$ Spec $k(x) \to X$. If $\mathcal{F}$ is any $\mathcal{O}_X$-module, write $\mathcal{F}(x) = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} k(x) = \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x$ a $k(x)$-vector space.

Definition. The **cotangent space** of X at $x \in X$ is the $k(x)$-vector space $T^*_{X/k,x} = \Omega_{X/k}(x)$.

Example. *If $X = \mathbb{A}^n_k$, then $\Omega_{X/k} = \widetilde{\Omega_{k[T_1,\cdots,T_n]/k}}$ is a free $\mathcal{O}_X$-module with basis $dT_1, \cdots, dT_n$. So for any (not necessarily closed) point $x \in X$, $T^*_{\mathbb{A}^k_n/k,x} = \bigoplus k(x) \cdot dT_i$.*

In scheme theory,"nilpotents" play the role of "infinitesimal" in calculus.

Definition. Let K be any field. The ring of **dual numbers** over K is $K[\varepsilon] = K[T]/(T^2)$.

The scheme Spec $K[\varepsilon]$ is not reduced: it has one point, and a canonical closed reduced subscheme Spec $K \hookrightarrow$ Spec $K[\varepsilon], K[\varepsilon] \to K$ given by $\varepsilon \mapsto 0$.

Definition. Let X be any k-scheme. The Zariski tangent space to X at $x \in X$ is the set $T_{X/k,x}$ consisting of k-scheme morphism $\theta : \operatorname{Spec} k(x)[\varepsilon] \to X$ whose restriction to $\operatorname{Spec} k(x) \subset \operatorname{Spec} k(x)[\varepsilon]$ is x.

Remark. *Intuitively, think of* $\operatorname{Spec} K[\varepsilon]$ *as a point plus a tangent direction.*

Theorem 2.7. *There is a canonical bijection:*

$$T_{X/k,x} \simeq Hom_{k(x)}(T^*_{X/k,x}, k(x)) = Hom_{\mathcal{O}_{X,x}}(\Omega_{X/k,x}, k(x))$$

This means T is the dual of T^*.

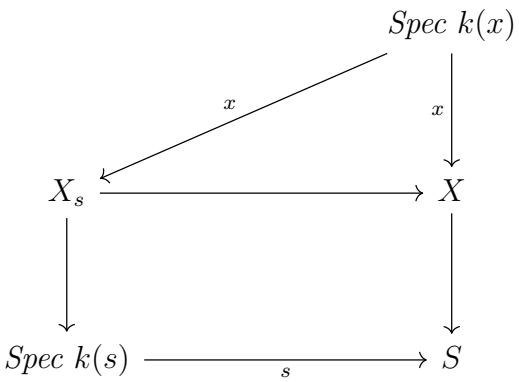
Proof. By localization, $\Omega_{X/k,x} = \Omega_{\mathcal{O}_{X,x}/k}$, and since $\operatorname{Spec} k(x)[\varepsilon]$ is a single point, every θ factors through $\operatorname{Spec} \mathcal{O}_{X,x} \to X$. So without loss of generality, $X = \operatorname{Spec}(A)$ for $(A, m, K = A/m)$ a local k-algebra, $x \in X$ the closed point. Then $T_{X/k,x}$ is the set of k-algebra local homomorphisms $\varphi : A \to K[\varepsilon]$ such that for any $a \in A$, $\varphi(a) \bmod \varepsilon = \overline{a}$. Any such φ is of the form $\varphi(a) = \overline{a} + D(a)\varepsilon$ for a map $D : A \to K$ k-linear. Now we check that φ is a homomorphism if and only if D is a k-derivation $A \to K$. So $\varphi \mapsto D$ is a bijection:

$$T_{X/k,x} \simeq \operatorname{Der}_k(A, K) = \operatorname{Hom}_A(\Omega_{A/k}, K)$$

In particular, the tangent space is a $k(x)$-vector space. □

Remark. *The base field doesn't play much role here. So, we can replace* $\operatorname{Spec} k$ *by any base scheme S, and define $T^*_{X/S,x} = \Omega_{X/S}(x)$, and $T_{X/S,x}$ be the collection of S-morphism θ.*

But generality is illusory: if $f(x) = s \in S$, $X_s = X \times_S \operatorname{Spec} k(s)$ is the fibre at s, then $T^*_{X/S,x} = T^*_{X_s/k(s),x}$ and $T_{X/S,x} = T_{X_s/x(s),x}$.



Proposition 2.8. *Let X be k-scheme, suppose $x \in X$ with $k(x) = k$. Then $\Omega_{X/k}(x) \cong (\mathfrak{m}_x/\mathfrak{m}_x^2)^*$ as k-vector spaces.*

Proof. As both sides depend only on $\mathcal{O}_{X,x}$, may assume $X = \operatorname{Spec} A$ for $(A, \mathfrak{m})$ local k-algebra with $A/\mathfrak{m} = k$. Now we consider the second exact sequence for $k \to A \to A/\mathfrak{m} = k$:

$$\mathfrak{m}/\mathfrak{m}^2 \to \Omega_{A/k} \otimes_A k = T^*_{X/k,x} \to \Omega_{k/k} = 0$$

So we only need to show the first map $f \bmod \mathfrak{m}^2 \mapsto df \otimes 1$ is injective. Equivalently: $\operatorname{Hom}_A(\Omega_{A/k}, k) = \operatorname{Der}_k(A, k) \to \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$ is surjective. If $D \in \operatorname{Der}_k(A, k)$ and $x, y \in m$, then $D(xy) = \overline{x}D(y) + \overline{y}D(x) = 0$. So $D(\mathfrak{m}^2) = 0$, so $\operatorname{Der}_k(A, k) = \operatorname{Der}_k(A/m^2, k)$. But $A = k \oplus m$ and $A/\mathfrak{m}^2 = k \oplus \mathfrak{m}/\mathfrak{m}^2$ with multiplication $(0, x)(b, y) = (ab, ay + bx)$. So $D \in \operatorname{Der}_k(A/\mathfrak{m}^2, k) \iff D$ is composite $A/\mathfrak{m}^2 \twoheadrightarrow \mathfrak{m}/\mathfrak{m}^2 \to k$. □

Remark. *If more generally $k(x)$ is a finite separable extension of k, then conclusion of the above theorem holds. This is because $\Omega_{k(x)/k} = 0$ and $A/\mathfrak{m}^2 \simeq k(x) \oplus \mathfrak{m}/\mathfrak{m}^2$.*

Example. *If $k = \overline{k}$, then for any closed point $x \in X : T^*_{X/k,x} = \mathfrak{m}_x/\mathfrak{m}_x^2$.*

Lemma 2.9. *Let A be a ring, and a homomorphism $\varphi : A^m \to A^n$ of A-modules. Then:*

$$\{\mathfrak{p} \in \text{Spec } A : \text{rank } (\varphi \otimes k(\mathfrak{p}) : k(\mathfrak{p})^m \to k(\mathfrak{p})^n) \leq r\}$$

is a closed subset of Spec A.

Proof. The set we are interested in is precisely the collection of $\mathfrak{p}$ such that all $(r+1)\times(r+1)$-minors of $\varphi \otimes k(\mathfrak{p})$ vanish. □

Proposition 2.10. *Let X be a locally noetherian scheme, $\mathcal{F}$ coherent $\mathcal{O}_X$ module. Then:*

1. *function $x \mapsto d(x) = \dim_{k(x)}(\mathcal{F}(x))$ is upper semicontinuous.*
2. *If X is reduced, d constant $\implies$ $\mathcal{F}$ is locally free.*

Proof. 1. Need to show that $Z_r = \{x \in X : d(x) \geq r\}$ is closed for any $r \in \mathbb{N}$. Can check locally, so we may assume $X = \text{Spec } A$ for A noetherian, $\mathcal{F} = \widetilde{M}$. We give a presentation $A^m \xrightarrow{\varphi} A^n \to M \to 0$. Then for any $x \in X$, then induced sequence:

$$k(x)^m \xrightarrow{\varphi \otimes k(x)} k(x)^n \to M \otimes_A k(x) \to 0$$

is exact. So $d(x) = n - \text{rank}(\varphi \otimes k(x)) \implies Z_r = \{x : \text{rank}(\varphi \otimes k(x)) \leq n - r\}$ is closed by Lemma.

2. (Reduced essential, consider $X = \text{Spec } k[\varepsilon]$). Let d be the common value of $d(x)$. Suppose first there exists a surjection $\pi : A^d \twoheadrightarrow M$ (we assume $X = \text{Spec } A, \mathcal{F} = \widetilde{M}$). By hypothesis $\pi \otimes k(\mathfrak{p}) : k(\mathfrak{p})^d \simeq M \otimes k(\mathfrak{p})$ for all $\mathfrak{p} \in \text{Spec } A$. So $(A/\mathfrak{p})^d \hookrightarrow M/\mathfrak{p}M$. So $\ker(\pi) \subset \cap_{\mathfrak{p}} \mathfrak{p}^d = 0$, since A is reduced. So in this case $M \simeq A^d$ is free.

   So in general case, suffices to show for any $\mathfrak{m} \subseteq A$ maximal, there exists $D(f) = \text{Spec } A_f$ containing $\mathfrak{m}$ with $\mathcal{F}|_{D(f)} = \widetilde{M_f}$ free.

   Let $k = A/\mathfrak{m}$, and pick $M \otimes_A k = \bigoplus_{1 \leq i \leq d} m_i \otimes 1, m_i \in M$. Put $M' = \sum Am_i \subset M$. So $M = M' + \mathfrak{m}M$. By Nakayama's lemma, there exists $f \in 1+\mathfrak{m}$ such that $f(M/M') = 0$. Then $(m_i)_{i=1}^d : A_f^d \twoheadrightarrow M'_f = M_f$. So by previous step, $M_f = A_f^d$ is free, and as $f \in 1 + \mathfrak{m}, \mathfrak{m} \in D(f)$.

□

Definition. Let X be an **integral** k-scheme of **finite type**, $d = \dim(X)$. Then for any $x \in X$, $\dim_{k(x)} T^*_{X/k,x} = \dim_{k(x)} T_{X/k,x} \geq d$. We say X is smooth at x if $\dim_{k(x)} T^*_{X/k,x} = d$. We say X is smooth over k if X is smooth everywhere.

Proof. Let $\eta \in X$ be the generic point. Then for any $x \in X$, $\dim T^*_{X/k,x} \geq \dim T^*_{X/k,\eta} = \dim_K(\Omega_{K/k})$ for $K = k(X)$ by upper semicontinuity. By Q6 of sheet, this is greater than the transcendence degree of the extension K/k. □

Theorem 2.11. *1. X^{sm} of smooth points of X is open.*

2. *A point $x \in X$ is smooth at $x \in X$ if and only if there is an open neighborhood U of x such that $\Omega_{X/k}|_U \cong \mathcal{O}_U^d$, for $d = \dim(X)$. So X is smooth iff $\Omega_{X/k}$ is locally free of rank d.*

3. *Let k'/k be any algebraic extension, and $p : X' = X \times_k \operatorname{Spec}(k') \to X$. Assume X' is also integral, $x' \in X', x = p(x')$. Then X' is smooth over k' at x' iff X is smooth over k at x. (Note p is surjective by going-up theorem, so X is smooth over k iff X' is smooth over k'.)*

Proof. 1. Apply Proposition 2.10 to $\Omega_{X/k}$, we see non-smooth points are closed.

2. Assume there exists $\Omega_{X/k}|_U \cong \mathcal{O}_U^d \implies T^*_{X/k,x} = k(x)^d$ trivially. Conversely, let $x \in V = X^{\mathrm{sm}}$. Then $\dim_{k(x)} \Omega_{X/k}(y) = d$ for all $y \in V$. Since X is reduced, this implies $\Omega_{X/k}|_V$ is locally free of rank d, by Proposition 2.10.

3. Differentials behave well under the base change: $\Omega_{X'/k'} = p^*\Omega_{X/k}$. So $T^*_{X'/k',x'} = T^*_{X/k,x} \otimes_{k(x)} k(x)$, so the dimensions are equal (and $\dim X = \dim X'$).

□

Theorem 2.12. *If X is a k-variety. Then:*

1. *X^{sm} is nonempty.*

2. *x is a smooth point $\iff$ $\Omega_{X/k}$ is free in a neighborhood of x.*

Proof. 1. We can base change to $\overline{k}$, and assume k is algebraically closed. Let η be the generic point, and $K = k(\eta)$ be the function field of X. By commutative algebra, we can find $k \subset K_0 = k(t_1, \cdots, t_d) \subset K$, for K_0/k purely transcendental of degree $d = \dim(X)$, and K/K_0 finite separable. Then $T^*_{X/k,\eta} = \Omega_{K/k} = \Omega_{K_0/k} \otimes_{K_0} K = \oplus K dt_i$. So $\eta \in X^{\mathrm{sm}}$.

2. We know x is smooth at generic point (since X^{sm} is nonempty open), $\dim T^*_{X/k,\eta} = d = \dim X$. Note we don't quote the first part since we don't want to perform the base change. We only need to show the direction $\impliedby$. Suppose $x \in U$ such that $\Omega_{X/k}|_U$ is free. We can spot its rank to be d since $\eta \in U$. It follows x is a smooth point from the above theorem.

□

Remark. *Morally speaking, x is smooth if "locally X looks like $\mathbb{A}^d$ in a neighborhood of x".*

Example. *Local freeness is not enough for checking smoothness in general. Consider $k = \mathbb{F}_p(t), X = \operatorname{Spec} K$, for $K = k(t^{\frac{1}{p}}) = k(s)$. Note X is integral of finite type over k with $\dim(X) = 0$. But we can compute $\Omega_{K/k} = K \cdot ds$, which is free but of "incorrect" rank. So X is not smooth over k (the problem is that K/k is inseparable).*

Theorem 2.13. *Let X be a variety over algebraically closed field k. Let $x \in X$ be a closed point (i.e $k(x) = k$). Then x is a smooth point $\iff$ $\mathcal{O}_{X,x}$ is regular: $\mathfrak{m}_x$ can be generated by d elements.*

Proof. This follows easily from Nakayama's lemma. □

If $k \neq \overline{k}$, this can be false (e.g. consider finite inseparable extensions of k).

Example. *Let $k = \mathbb{F}_p(a), p > 2$. Let $X = \operatorname{Spec} k[u,v]/(v^2 - u^p + a) = \operatorname{Spec} A$. It is geometrically integral, since the defining polynomial is irreducible over $\overline{k}$. Let $x = \mathfrak{p} = (v, u^p - a) = (v) \subset A$, and $A/\mathfrak{p} = k[u]/(u^p - a) = k(a^{\frac{1}{p}})$. So x is a closed point. Note m_x is generated by a single element v, so $\mathcal{O}_{X,x}$ is regular. However:*

$$\Omega_{A/k} = \frac{Adu + Adv}{A \cdot vdv} = A \cdot du + \frac{A}{vA} \cdot dv$$

So $\dim T_{X/k,y} = 1, \forall y \neq x, \dim T_{X/k,x} = 2$. So X isn't smooth at x. The problem is: over $\overline{k}$, the equation of X becomes $v^2 = (u - a^{\frac{1}{p}})^p$.

Remark. *Let X be an integral k-scheme of finite type over k, $x \in X$. Then x is a smooth point $\iff$ there exists open affine neighborhood $U = \operatorname{Spec} A$ of X, with $A = k[t_1, \cdots, t_{m+d}]/(g_1, \cdots, g_m)$ ($d = \dim X$) such that the matrix:*

$$(\frac{\partial g_i}{\partial t_j}(x))_{1 \leq i \leq m, 1 \leq j \leq m+d}$$

has rank m. Given such a presentation, then the cotangent space is the cokerenal of the map given by this matrix.

Digression on Finiteness and Related Conditions

Let X be a scheme. Recall X is quasicompact $\iff$ X is finite union of open affines.

Example. *Here are a few non-examples. $X = \bigsqcup \operatorname{Spec} k$, $\mathbb{A}^\infty_k \backslash \{origin\}$*

X is separated $\implies$ intersection of any two open affines is affine.

Definition. We call X is **quasi-separated**, if the intersection of any two open affines is quasi-compact.

Example. *non-example: $\mathbb{A}^\infty \cup \mathbb{A}^\infty$ along the complement of the origin.*

Exercise. Find scheme $X = U \cup V$ of open affines, such that $U \cap V = \bigsqcup_{\mathbb{N}} \operatorname{Spec} k$.

If X is qcqs(quasicopmact and quasiseparated). Then $f : X \to Y$ is quasicompact if for any open affine $U \subset Y$ such that $f^{-1}(U)$ is quasicompact; we call f locally of finite type, if for any $x \in X$, there exists open affine $x \in U = \operatorname{Spec} B \subset X$, $f(U) \subset V = \operatorname{Spec} A \subset X$ with B an A-algebra of finite type; we call f finite type if locally of finite type and quasicompact.

We call f is locally of finite presentation: same with $B = A[t_1, \cdots, t_n]/I$, for I finitely generated ideal; f of finite presentation if moreover it is quasicompact and quasiseparated.

We next list several finiteness properties on $\mathcal{O}_X$-modules:

1. quasi-coherent

2. of finite type: locally there exists a surjection $\mathcal{O}_U^n \twoheadrightarrow \mathcal{F}$

3. for X locally Noetherian, $\mathcal{F}$ is coherent $\iff$ on opens it is isomorphic to $\widetilde{M}$ for some finite module M $\iff$ quasi-coherent and of finite type. For non-noetherian schemes, there is a definition (already in Serre's FAC): it needn't be the case for such a scheme that $\mathcal{O}_X$ is coherent.

4. Locally free: each point $x \in X$ has an open neighborhood U such that $\mathcal{F}_U$ is free. Note that if $\mathcal{F}$ locally free, then each stalk $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$-module for any $x \in X$. But stalkwise free doesn't imply locally free: take $X = \operatorname{Spec} \mathbb{Z}, M = \{\frac{a}{b} \in \mathbb{Q} | b \text{ square free}\}$. We compute $M_{(p)} = \frac{1}{p}\mathbb{Z}_{(p)}$ is free. But $M \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{n}]$ isn't free for any $n \geq 1 \implies \widetilde{M}$ not locally free.

3 Flatness and Smooth Morphisms

Definition. Let A be a ring. We call an A-module M is flat if tensoring with M is left exact. We say an A-algebra B is flat if it is flat as an A-module.

Lemma 3.1. *We have the following properties of flat modules:*

1. *Free modules are flat.*

2. *M is flat $\iff$ for any finitely generated ideal $I \subset A$, the map $M \otimes_A I \to M$ is injective (i.e. $M \otimes I \simeq IM \subset M$).*

3. *M is flat $\iff$ tensoring with M is an exact functor.*

4. *M is flat $\iff$ for any $m_1, \cdots, m_n \in M$, $a_1, \cdots, a_n \in A$ such that $\sum_{i=1}^n a_i m_i = 0$, there exists $e_1, \cdots e_r \in M$ and $b_{ij} \in A$ with $m_i = \sum b_{ij} e_j$ and $\forall j, \sum_i a_i b_{ij} = 0$. (M free, can take (e_i) part of the basis).*

5. *Direct limits of flat modules are flat.*

6. *$A \to B$ ring homomorphism, M a flat A-module $\implies$ $M \otimes_A B$ is a flat B-module.*

7. *M is A-flat $\iff$ $M_{\mathfrak{p}}$ is $A_{\mathfrak{p}}$ flat for each prime ideal $\mathfrak{p}$.*

8. *B flat A-algebra, M flat B-module $\implies$ M is flat as an A-module.*

9. *A a field, then every A-module is flat; if A is a PID, then A-flat is equivalent to A-torsion free.*

10. *A Noetherian, M finite A-module. Then M is flat $\iff$ M is projective $\iff$ $\widetilde{M}$ is locally free on* $\operatorname{Spec} A$ *$\iff$ $M_{\mathfrak{p}}$ is free for each $\mathfrak{p} \in \operatorname{Spec} A$*

Remark. *Indeed, filtered colimits of flat modules are flat, but not necessarily for arbitrary colimit. For example, the diagram $\mathbb{Z} \underset{2}{\overset{0}{\rightrightarrows}} \mathbb{Z}$ has colimit $\mathbb{Z}/2\mathbb{Z}$, which is not flat.*

Proposition 3.2. *Let* $\cdots \to K^{n-1} \xrightarrow{d^{n-1}} K^n \xrightarrow{d^n} K^{n+1} \to \cdots$ *be a complex of* A*-modules. Let* M *be an* A*-module, then there is a natural homomorphism* $H^n(K^\bullet) \otimes_A M \to H^n(K^\bullet \otimes_A M)$*, which is an isomorphismfor* M *flat.*

Proof. We have:

$$0 \to \ker(d^n) \to K^n \xrightarrow{d^n} K^{n+1}$$

this is exact. So:

$$\ker(d^n) \otimes_A M \to \ker(d^n \otimes \mathrm{id}_M)$$

by functoriality of tensor product. On the other hand, we have:

$$K^{n-1} \xrightarrow{d^{n-1}} \ker(d^n) \to H^n(K^\bullet) \to 0$$

is exact. Hence so are the rows of:

$$\begin{array}{ccccccc}
K^{n-1} \otimes M & \xrightarrow{d^{n-1}} & \ker(d^n) \otimes M & \longrightarrow & H^n(K^\bullet) \otimes M & \longrightarrow & 0 \\
\Big\| & & \downarrow & & \downarrow{\scriptstyle \psi} & & \downarrow \\
K^{n-1} \otimes M & \longrightarrow & \ker(d^n \otimes \mathrm{id}_M) & \longrightarrow & H^n(K^\bullet \otimes M) & \longrightarrow & 0
\end{array}$$

defining ψ . If M is flat, then the second vertical map is an isomorphism, so is ψ . □

Definition. Let $f : X \to Y$ be a morphism of schemes, $x \in X$. Then f is flat at x if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$-algebra. We call f flat if it is flat at all $x \in X$.

A quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ is flat over Y if $\mathcal{F}_x$ is flat as a $\mathcal{O}_{Y,f(x)}$ module for any $x \in X$. (So f flat $\iff$ $\mathcal{O}_X$ is flat over Y).

Remark. *1.* Spec $B \to$ Spec A *is flat* $\iff$ B *is* A*-flat.*

2. *open immersions are flat, closed immersions are rarely flat.*

3. *Let* X, Y *be* k*-schemes. Then* $X \times_k Y \to Y$ *is flat. (*$A \otimes_k B$ *flat over* B *as* A *is* k*-flat)*

4. *Morally, flat morphism = "continuously varing family" (read section 9, Chapter 3 of Hartshorne).*

Proposition 3.3. *If* $f : X \to Y$ *is a morphism of integral scheme of finite type over a field. If* f *is flat, then every* $y \in Y$ *and every* $Z \subset f^{-1}(y)$ *irreducible component of the fibre (this is a scheme of finite type over* Spec $k(y)$*, not necessarily integral),* $\dim Z + \dim Y = \dim X$.

If Y is regular, the converse is true:

Theorem 3.4. *Let* $f : X \to Y$ *be a morphism of integral schemes of finite type over a field, with* Y *regular. Suppose* X *is also regular [this can be weakened a lot], and for any* $y \in Y$ *closed,* $Z \subset f^{-1}(y)$ *irreducible,* $\dim X = \dim Z + \dim Y \implies f$ *is flat.*

Remark. *The requirement for Y to be regular is essential. Let $Y = \operatorname{Spec} k[u,v]/(u^3 - v^2)$ be a singular cubic curve, and X be its normalization: $X = \operatorname{Spec} k[t] = \mathbb{A}^1$, $u = t^2, v = t^3$. The map $f : X \to Y$ is not flat: if so, then the $k[t]$ will be a locally free module over $k[u,v]/(u^3 - v^2)$, contradiction.*

However, we can just assume X is locally a complete intersection: locally of form

$$\operatorname{Spec} \frac{k[t_1, \cdots, t_{m+\dim X}]}{(f_1, \cdots, f_m)}$$

This is known as "Miracle Flatness".

Example. *Let $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ given in coordinate $(u,v) \mapsto (u, uv)$. If $u \neq 0$, the fibre above (u, uv) is a single point. But $f^{-1}(0,0) = \{u = 0\}$ is a line. So f is not flat. (blowup)*

Altough fibre dimensions is constant in a flat family, other things can vary. Consider $Y = \mathbb{A}^1_k = \operatorname{Spec} k[t] \xleftarrow{f} X = \operatorname{Spec} k[x,y,t]/(xy - t)$ or $\operatorname{Spec} k[x,y,t]/(x^2 - ty)$. Look at the fibre over the origin $t = 0$.

Example. *$X \xrightarrow{f} Y = \operatorname{Spec} A$, A PID (or Dedekind domain). f is flat $\iff$ all local rings $\mathcal{O}_{X,x}$ are A-torsion free.*

If X is integral with generic point η, then $\mathcal{O}_{X,\eta}$ is A-torsion free, we see $f(\eta)$ is the generic point of $\operatorname{Spec} A$*. So f is dominant. If moreover X is proper, then f is surjective.*

In general, every irreducible component of X dominates Y.

Proposition 3.5. *Let $f : X \to Y$ be flat and locally of finite presentation (e.g. of finite type with Y noetherian). Then f is open.*

Remark. *Note* $\operatorname{Spec} \mathbb{Q} \to \operatorname{Spec} \mathbb{Z}$ *is flat but not open.*

For flat morphisms, read Hartshorne III.9, especially example III.9.8.4 (don't use the first edition!).

Nicest sort of flat family is a family with smooth fibres.

Definition. Let $f : X \to Y$ locally of finite presentation (e.g. for Y noetherian, f of finite type). Let $x \in X, y = f(x) \in Y$. Then f is smooth at x (of relative dimension d) if:

1. f is flat at x.
2. there exists an open neighborhood $x \in U \subset X_y = f^{-1}(y)$ such that U is smooth at x over $k(y)$, of dimension d.

We say f is smooth if it's smooth at every point. This is just saying f is flat and each connected component of every fibre are smooth over $k(y)$.

Example. *Let Y, Z be varieties over k, Z smooth over k. The $X = Y \times Z \to Z$ is smooth.*

Theorem 3.6. *Let $f : X \to Y$ be a morphism of finite presentation. The following are equivalent*

1. *f is smooth*

2. *f is flat, $\Omega_{X/Y}$ is locally free of the fibre dimension.*

3. *for all $x \in X$, there exists an open affine neighborhood $U = \operatorname{Spec} B$ of x, $f(U) \subset \operatorname{Spec} A \subset Y$, with $B = A[t_1, \cdots, t_{m+d}]/(f_1, \cdots, f_m)$ such that the matrix*

$$(\frac{\partial f_i}{\partial t_j}(x))_{i,j} \in M_{m,m+d}(k(x))$$

has rank m.

The second condition says that for every connected component $Y' \subset Y$, and $\Omega_{X/Y}$ is locally free of rank d on $f^{-1}(Y')$, where d is the dimension of every irreducible component of $f^{-1}(y'), y' \in Y'$.

If $d = 0$, smooth morphisms are intuitively a local isomorphism.

Definition. We call f étale if it's smooth of dimension 0. So locally of form $\operatorname{Spec} B \to \operatorname{Spec} A$, where $B = A[t_1, \cdots, t_m]/(f_1, \cdots, f_m)$ with the Jacobian matrix $\frac{\partial f_i}{\partial t_j}$ invertible. (c.f. implicit function theorem)

Example. *Consider $f : \operatorname{Spec} k[u, v]/(u - v^2) \to \operatorname{Spec} k[v]$, $v \mapsto v^2 = u$. Away from $u = v = 0$, f is étale.*

Remark. *For $f : X \to Y$ locally of finite presentation, f is étale if it's flat and unramified: $\Omega_{X/Y} = 0$.*

4 Sheaf Cohomology

In this section, all schemes will be separated and noetherian, unless otherwise state. In particular, every scheme will have a finite open affine cover with intersections of these affines also affine

4.1 Some Homological Algebra

Let $(A^\bullet, d^\bullet)$ is a complex of R-modules or sheaves. Mostly we assume the complex starts with degree zero. We can thus define the cohomology, and we can define an graded object:

$$H^*(A^\bullet) = \bigoplus_{n \neq 0} H^n((A^\bullet))$$

We can also define the notion of the morphism between complexes, which naturally induces a map between cohomology.

A basic fact is that if

$$0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$$

there is an induced long exact sequence (functorial w.r.t exact sequence):

$$0 \to H^0(A) \to H^0(B) \to H^0(C) \to H^1(A) \to \cdots$$

Also recall that chain homotopy gives same cohomology.

In particular, if we get $h^p : A^p \to A^{p-1}$ such that $h \circ d + d \circ h = \mathrm{id}_A$, then: $H^p(\mathrm{id}_A) = 0 \implies H^p(A) = 0, \forall p$. If such an exact h exist, say the complex $A^\bullet$ is null-homotopic. So null-homotopy implies vanishing of all cohomology.

For vector spaces over a field, the converse is true: if $H^*(A^\bullet) = 0$, then by linear algebra, we can write:

$$A^\bullet = [B^0 = A^0 \to B^0 \bigoplus B^1 = A^1 \to B^1 \bigoplus B^2 = A^2 \to \cdots]$$

with the obvious maps. Then the obvious map $A^p \to B^{p-1} \to A^{p-1}$ is a null-homotopy. This is not true for abelian groups, consider $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$.

Remark. *Work with R-modules. For $p \geq 0$, let $R[-p]$ be the shifted complex: R in degree p and 0 elsewhere. Then for any complex $A^\bullet$ of R-modules, then:*

$$H^p(A^\bullet) = \{\textit{homotopy classes of morphisms of complexes } R[-p] \to A^\bullet\}$$

Special case of the long exact sequence of cohomology(i.e. the Snake Lemma): Suppose we have commutative diagram:

$$\begin{array}{ccccccccc} & & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & & \end{array}$$

Then, we have a long exact sequence:

$$\ker f \to \ker g \to \ker h \to \operatorname{coker} f \to \operatorname{coker} g \to \operatorname{coker} h$$

Tensor Product of Complexes

Let $A^\bullet, B^\bullet$ be complexes of R-modules. We can define the tensor product complex $(A^\bullet \otimes_R B^\bullet, d_{A\otimes B})$ as following:

1. in degree n it is $\bigoplus_{p+q=n} A^p \otimes B^q$.
2. differential is $d : A^p \otimes B^q \to A^{p+1} \otimes B^q \bigoplus A^p \otimes B^{q+1}$, given by $d(a \otimes b) = da \otimes b + (-1)^{|a|} a \otimes db$.

Theorem 4.1 (Naive Kunneth Formula)**.** *Let $R = k$ be a field. Then*

$$H^n(A \otimes B) = \bigoplus_{p+q=n} H^p(A) \otimes_k H^q(B).$$

4.2 Sheaf Cohomology

Recall: Let $\mathcal{F}$ be a sheaf of abelian groups on topological space X. there are induced sheaf cohomology groups: $H^p(X, \mathcal{F}), p \geq 0$ such that:

1. $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$
2. For any short exact sequence of sheaves $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$, there exists a long exact sequence of cohomology:

$$0 \to H^0(X, \mathcal{F}_1) \to H^1(X, \mathcal{F}_2) \to H^0(X, \mathcal{F}_3) \to H^1(X, \mathcal{F}_1) \to H^1(X, \mathcal{F}_2) \to \cdots$$

They are defined using injective resolutions, but can be computed in various other ways:

1. Recall a sheaf $\mathcal{G}$ is flasque if restriction maps are all surjective. Then if we have a long exact sequence
$$0 \to \mathcal{F} \to G^0 \to G^1 \to \cdots$$
with each G^p flasque, then $H^p(X, \mathcal{F}) = H^p(\Gamma(X, \mathcal{G}))$. In particular, if $\mathcal{F}$ is flasque, then all higher cohomology vanishes.
2. Say sheaf $\mathcal{G}$ is acyclic on X, if for any $p \geq 0$, $H^p(X, \mathcal{G}) = 0$. Then if
$$0 \to \mathcal{F} \to G^0 \to G^1 \to \cdots$$
with each $\mathcal{G}^p$ acyclic, then $H^p(X, \mathcal{F}) = H^p(\Gamma(X, \mathcal{G}))$.

Every sheaf $\mathcal{F}$ has a canonical resolution by flasque sheaves (Godement resolution):

$$G\mathcal{F} = \prod_{x \in X} i_{x,*}(\mathcal{F}_x), \quad i_x : \{x\} \to X.$$

There is a canonical injection $\mathcal{F} \hookrightarrow G\mathcal{F}$, and $G\mathcal{F}$ is flasque by constrution: $G\mathcal{F}(U = \prod_{x \in U} \mathcal{F}_x)$. We can thus build a complex:

$$0 \to \mathcal{F} \xrightarrow{a_0} \mathcal{G}^0 = G\mathcal{F} \xrightarrow{a_1} G(\text{coker } a_0) = \mathcal{G}_1 \xrightarrow{a_2} \mathcal{G}_2 \to \cdots$$

Sheaf cohomology is functorial: suppose we have continuous map $f : X \to Y$, sheaves $\mathcal{F}, \mathcal{G}$ on X, Y, and $\varphi : \mathcal{G} \to f_*\mathcal{F} \iff f^*\mathcal{G} \to \mathcal{F}$. There is an induced map $H^p(Y, \mathcal{G}) \to H^p(X, \mathcal{F})$.

Proposition 4.2. *Suppose $i : Y \hookrightarrow X$ is closed subset, $\mathcal{F}$ is abelian sheaf onf Y. Then for any p,*

$$H^p(Y, \mathcal{F}) = H^p(X, i_*\mathcal{F})$$

Proof. Choose a flasque resolution of F on Y:

$$0 \to \mathcal{F} \to \mathcal{G}^0 \to \mathcal{G}^1 \to \cdots$$

So $H^*(Y, \mathcal{F}) = H^*(\Gamma(Y, G^\bullet))$. By the definition $i_*\mathcal{G}^n$ is a flasque sheaf on X, so:

$$H^p(X, i_*\mathcal{F}) = H^p(\Gamma(X, i_*\mathcal{G}^\bullet)) = H^p(\Gamma(Y, \mathcal{G}^\bullet)) = H^p(Y, \mathcal{F})$$

. □

Theorem 4.3 (Mayer-Vietoris Sequence)**.** *Let $U, V \subseteq X$ open with $U \cup V = X$. Let $\mathcal{F}$ be a sheaf on X. Then there is a long exact sequence:*

$$0 \to H^0(X, \mathcal{F}) \to H^0(U, \mathcal{F}|_U) \oplus H^0(V, \mathcal{F}|_V) \to H^0(U \cap V, \mathcal{F}|_{U\cap V}) \to H^1(X, \mathcal{F}) \to \cdots$$

Proof. Pick $0 \to \mathcal{F} \to \mathcal{G}^0 \to \mathcal{G}^1 \cdots$ a flasque resolution. Then we have the short exact sequence:

$$0 \to \Gamma(X, \mathcal{G}^\bullet) \to \Gamma(U, \mathcal{G}^\bullet) \oplus \Gamma(V, \mathcal{G}^\bullet) \to \Gamma(U \cap V, \mathcal{G}^\bullet) \to 0$$

It is left exact by the sheaf axioms, and is right exact since $\mathcal{G}^\bullet$ are flasque. The result follows by snake lemma. □

Now assume X is a scheme (separated and noetherian). For a sheaf $\mathcal{F}$ on X, we have $H^*(X, \mathcal{F})$.

Theorem 4.4. *1. If $\dim X = d$, then $H^p(X, \mathcal{F}) = 0$ for any $p > d$, $\mathcal{F}$ any sheaf.*

2. For quasicoherent O_X-modules, we can compute cohomology as Čech Cohomology.

3. More generally, if $\mathcal{F}$ has no cohomology on each U_I, then the Čech cohomology and the sheaf cohomology agree.

Lemma 4.5. *Suppose $j : V \hookrightarrow X$ open affine. Let $\mathcal{F}$ be a quasicoherent $\mathcal{O}_V$-module, then: $j_*\mathcal{F}$ is quasi-coherent and acyclic.*

Proof. X is separated, so j is affine. So $j_*\mathcal{F}$ is quasicoherent, and we can compute its Čech cohomology:

$$\check{C}^p((U_i), j_*\mathcal{F}) = \prod_{i_0 < \cdots < i_p} (U_{i_0} \cap \cdots U_{i_p} \cap V) = \check{C}^p((U_i'), \mathcal{F})$$

where $U_i' = V \cap U_i$. So $\check{H}^p((U_i, j_*\mathcal{F})) = \check{H}^p(U_i', \mathcal{F}) = 0$ for $p > 0$. □

Sheaf Version of Čech Cohomology

For $V \subset X$ open, define:

$$\underline{\check{C}}^p((U_i), \mathcal{F})(V) = \check{C}^p(U_i \cap V, \mathcal{F}|_V)$$

Note it is indeed a sheaf as:

$$\underline{\check{C}}^p((U_i), \mathcal{F}) = \prod_{\#I = p+1} j_{I*}(\mathcal{F}|_{U_I}) \quad j_I : U_I \hookrightarrow X$$

Then define Čech differentials as before, this gives a complex of sheaves:

$$0 \to \mathcal{F} \to \underline{\check{C}}^0((U_i), \mathcal{F}) \to \underline{\check{C}}^1((U_i), \mathcal{F}) \to \cdots \qquad (\star)$$

Proposition 4.6. *Assume (U_i) is an affine cover, $\mathcal{F}$ quasi-coherent. Then $\star$ is an exact sequence of quasi-coherent sheaves, and $H^n(X, \underline{\check{C}}^p((U_i), \mathcal{F})) = \begin{cases} \check{C}^p((U_i), \mathcal{F}), & \text{if } n = 0 \\ 0, & \text{if } n > 0 \end{cases}$*

Proof. Apply previous lemma to $j_{I*}(\mathcal{F}|_{U_I})$. So the sheaves $\underline{\check{C}}^p((U_i),\mathcal{F})$ are acyclic for all p. $H^0(X,\underline{\check{C}}^p) = \check{C}^p((U_i),\mathcal{F})$. It remains to check the exactness of the sequence. Let $V \subset X$ be open affine, then the sections over V of $\star$ are precisely the Čech complex computing the cohomology of $\mathcal{F}|_V$ over V. Since V is affine, we see the complex is exact. □

Theorem 4.7 (Kunneth formula)**.** *Let X, Y be separated noetherian k-schemes. Let $\mathcal{F}, \mathcal{G}$ be quasi-coherent $\mathcal{O}_X, \mathcal{O}_Y$-modules. Then*

$$H^*(X \times_k Y, \mathcal{F} \boxtimes \mathcal{G}) = \bigoplus_{p+q=n} H^q(X,\mathcal{F}) \otimes H^q(Y,\mathcal{G})$$

Lemma 4.8. *1. Let A, B be k-algebras, M be an A-module, N be a B-module. Then: $\widetilde{M} \boxtimes \widetilde{N} \cong \widetilde{M \otimes_k N}$*

2. If $\mathcal{F}, \mathcal{G}$ are in theorem, and $\mathcal{F} \to \mathcal{K}^\bullet, \mathcal{G} \to \mathcal{L}^\bullet$ resolutions by quasicoherent sheaves. Then we get a resolution:

$$\mathcal{F} \boxtimes \mathcal{G} \to \mathcal{K}^\bullet \boxtimes \mathcal{L}^\bullet$$

Proof. 1. This is trivial.

2. It is enough to check on an open affine covering. So we may assume X, Y affine and $\mathcal{F} = \widetilde{M}, \mathcal{G} = \widetilde{N}$. But then we conclude by naive Kunneth formula.

□

Proof. (of Kunneth formula) Take affine covers $\underline{U} = (U_i), \underline{(V)} = (V_j)$ for X and Y. Then we have the Čech resolutions:

$$0 \to \mathcal{F} \to \underline{\check{C}}^0((U_i),\mathcal{F}) \to \underline{\check{C}}^1((U_i),\mathcal{F}) \to \cdots$$

$$0 \to \mathcal{G} \to \underline{\check{C}}^0((V_i),\mathcal{G}) \to \underline{\check{C}}^1((V_i),\mathcal{G}) \to \cdots$$

By the lemma, we get a resolution:

$$0 \to \mathcal{F} \boxtimes \mathcal{G} \to \mathcal{K}^\bullet := \underline{\check{C}}^\bullet((U_i),\mathcal{F}) \boxtimes \underline{\check{C}}^\bullet((V_i),\mathcal{G})$$

We compute $\mathcal{K}^n = \bigoplus_{p+q=n} \prod_{\#I=p+1, \#J=q+1} \mathcal{K}_{I,J}$, where $K_{I,J} = (j_I \times j_J)_\star(\mathcal{F} \boxtimes \mathcal{G})|_{U_I \times V_J}$, and $j_I \times j_J : U_I \times V_J \hookrightarrow X \times Y$ is inclusion of an open affine. From previous lemma we see that $\mathcal{K}^n$ is acyclic. So $H^*(X \times Y, \mathcal{F} \boxtimes \mathcal{G}) = H^*(\Gamma(X \times Y, K^\bullet))$.

But $\Gamma(X \times Y, \mathcal{K}_{I,J}) = \mathcal{F}(U_I) \otimes \mathcal{G}(V_J)$, so:

$$\Gamma(X \times Y, \mathcal{K}^\bullet) = \check{(}C)^\bullet(\underline{U},\mathcal{F}) \otimes \check{(}C)^\bullet(\underline{V},\mathcal{G})$$

whose cohomology by naive Kunneth formula, is indeed $\bigoplus_{p+q=n} H^q(X,\mathcal{F}) \otimes H^q(Y,\mathcal{G})$ in degree n. □

This gives rise to the **cup product** in cohomology as follows:

Let $Y = X$, diagonal $\Delta : X \to X \times X$. For $\mathcal{F}, \mathcal{G}$ quasi-coherent on X. Then $\Delta^*(\mathcal{F} \boxtimes \mathcal{G}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$. Cup product is the following composite map:

$$H^p(X, \mathcal{F}) \otimes H^q(X, \mathcal{G}) \hookrightarrow H^{p+q}(X \times X, \mathcal{F} \boxtimes \mathcal{G}) \xrightarrow{\Delta^*} H^{p+q}(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$$

If we take $\mathcal{F} = \mathcal{G} = \mathcal{O}_X$, we get:

$$H^p(X, \mathcal{O}_X) \otimes H^q(X, \mathcal{O}_X) \to H^{p+q}(X, \mathcal{O}_X)$$

and this makes $H^*(X, \mathcal{O}_X)$ a graded k-algebra. One can show this is graded commutative.

Cohomology of Projective Space, Hilbert Polynomials

Let X be noetherian separated, $\mathcal{F}$ coherent $\mathcal{O}_X$-module.

Definition. The support of $\mathcal{F}$ is

$$\operatorname{supp}(\mathcal{F}) = \{x \in X | \mathcal{F}_x \neq 0\}$$

This is closed in X, since if $X = \operatorname{Spec} A, \mathcal{F} = \widetilde{M}$, then

$$\operatorname{supp}(\mathcal{F}) = \{\mathfrak{p} | M_{\mathfrak{p}} \neq 0\} = V(\operatorname{ann}(M))$$

More precisely, there exists a closed subscheme $i : Z \hookrightarrow X$ whose points are precisely $\operatorname{supp}(\mathcal{F})$ not necessarily reduced, such that $\mathcal{F} = i_* \mathcal{G}$ for a coherent $\mathcal{O}_Z$-module $\mathcal{G}$, and Z is "as small as possible".

Example. *Let* $X = \operatorname{Spec} A, \mathcal{F} = \widetilde{M}, Z = \operatorname{Spec} \frac{A}{\operatorname{ann}(M)}$.

Theorem 4.9. *Noetherian ring A, $f : X \to \operatorname{Spec} A$ proper (so X is noetherian, separated), $\mathcal{F}$ a coherent $\mathcal{O}_X$-module, then $H^p(X, \mathcal{F})$ is a finite A-module. In particular, for a proper k-scheme X, $\dim_k H^p(X, \mathcal{F}) < \infty$.*

Definition. For X/k proper, $\mathcal{F}$ coherent. Then the **Euler Characteristic** of $\mathcal{F}$ is

$$\chi(X, \mathcal{F}) = \sum_{p \geq 0} (-1)^p \dim_k H^p(X, \mathcal{F}) \in \mathbb{Z}.$$

Remark. *One can easily sees that Euler Characteristic is additive.*

Let $\mathbb{P}^N = \mathbb{P}^N_k$. Recall if $U_j = \operatorname{Spec} k[\frac{T_0}{T_j}, \cdots, \frac{T_N}{T_j}]$, then $\mathcal{O}(n)$ has section

$$\left\{ \frac{f}{T_j^d} | f \text{ homogeneous of degree } n + d \right\}$$

Its Cohomology is:

$$H^p(\mathbb{P}^N, \mathcal{O}(n)) = \begin{cases} k[T_0, \cdots, T_N]_{\deg = n} & \text{if } p = 0, n \geq 0; \\ k[T_0, \cdots, T_N]^\vee_{\deg = -N-n-1} & \text{if } p = N, n \leq -N-1; \\ 0 & \text{otherwise.} \end{cases}$$

So $\chi(\mathbb{P}^N, \mathcal{O}(n)) = P(n)$, where $P(t) = \frac{(t+N)\cdots(t+1)}{N!} \in \mathbb{Q}[t]$.

Theorem 4.10. *Let $\mathcal{F}$ be any coherent sheaf on $\mathbb{P}^N$, then there eixsts a polynomial $P(\mathcal{F}, t) \in \mathbb{Q}[t]$, the **Hilbert polynomial** of $\mathcal{F}$, such that for all $n \in \mathbb{Z}$,*

$$\chi(\mathbb{P}^N, \mathcal{F} \otimes \mathcal{O}(n)) = P(\mathcal{F}, n)$$

and $\deg P(\mathcal{F}, t) = \dim(\operatorname{supp}(\mathcal{F}))$.

More generally, let X be a projective scheme over k, $i : X \hookrightarrow \mathbb{P}^N$. Define $\mathcal{O}_X(n) = i^*\mathcal{O}_{\mathbb{P}^n}(n)$. Conversely, for any k-scheme X and line bundle $\mathcal{L}$, we say $\mathcal{L}$ **very ample** if there is a closed immersion $i : X \hookrightarrow \mathbb{P}^n$ with $\mathcal{L} \simeq \mathcal{O}_X(1)$.

Let X be projective, $\mathcal{F}$ coherent on X, set $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$. We know

$$H^*(X, \mathcal{F}(n)) \cong H^*(\mathbb{P}^N, i_*\mathcal{F}(n))$$

Define $P(X, \mathcal{F}, t) \coloneqq P(i_*\mathcal{F}, t)$, then we know $P(X, \mathcal{F}, n) = \chi(X, \mathcal{F}(n))$

Remark. *Suppose $\mathcal{L}$ is very ample line bundle. Then if $s_0, \cdots, s_N$ is a basis for $\Gamma(X, \mathcal{L})$, it determines a closed immersion $i : X \hookrightarrow \mathbb{P}^N$ where $s_j = i^*(T_j)$, where $T_j \in \Gamma(\mathbb{P}^N, \mathcal{O}(1))$.*

Theorem 4.11 (Criterion for Ampleness)**.** *Let X/k are proper, $\mathcal{L}$ line bundle on X. TFAE:*

1. *for some $r \geq 1$, $\mathcal{L}^{\otimes r}$ is very ample (this must hold for all r large enough)*
2. *For any coherent $\mathcal{O}_X$-module $\mathcal{F}$, there exists $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is generated by global sections.*
3. *(Serre's Criterion) For any coherent $\mathcal{O}_X$-module $\mathcal{F}$, there exists n_0 such that for all $n \geq n_0$ and $p > 0$*
$$H^p(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$$

If these hold, say $\mathcal{L}$ ample.

Example. *Let X be an elliptic curve over k, $P \in X$ be a k-rational point. Let $\mathcal{L} = \mathcal{O}_X(P)$. We know $\mathcal{L}^{\otimes 3}$ is very ample. So $\mathcal{L}$ is ample, but not very ample.*

In general, for a smooth proper curve X, $\mathcal{L}$ is ample $\iff$ $\mathcal{L} \simeq \mathcal{O}_X(D)$ for $deg(D) > 0$.

Proposition 4.12. *Let $X \subset \mathbb{P}^N_k$ be integral, $\dim X = d$, η generic point, $\mathcal{F}$ a coherent $\mathcal{O}_X$-module. Then*

$$P(X, \mathcal{F}, t) = \dim_{k(X)} \mathcal{F}_\eta \cdot P(X, \mathcal{O}_X, t) + (\text{terms of degree} < d)$$

Note $P(X, \mathcal{O}_X, t)$ has degree $\dim \operatorname{supp}(\mathcal{O}_X) = d$.

Proof. Write $e_1, \cdots e_r$ be a $k(X)$-basis for $\mathcal{F}_\eta, r \geq 0$. Then there exists n such that $e_1, \cdots, e_r$ extend to global sections of $\mathcal{F}(n)$. So we have an exact sequence

$$0 \to \mathcal{O}_X^r \xrightarrow{(e_1, \cdots, e_r)} \mathcal{F}(n) \to \mathcal{G} \to 0$$

where $\mathcal{G}_\eta = 0$. So $\dim \operatorname{supp}(\mathcal{G}) < d$. Hence

$$\begin{aligned} P(X, \mathcal{F}, t + n) = P(X, \mathcal{F}(n), t) &= P(X, \mathcal{O}_X^{\oplus r}, t) + P(X, \mathcal{G}, t) \\ &= rP(X, \mathcal{O}_X, t) + (\text{degree} < d \text{ terms}) \end{aligned}$$

□

Cohomology and Base Change

Let $f : X \to Y = \operatorname{Spec} A$ noetherian separated, $\mathcal{F}$ be a quasicoherent $\mathcal{O}_X$-module. Consider $A \to B$, an A-algebra. Consider the base change:

$$\begin{array}{ccc} X_B = X \times_A \operatorname{Spec}(B) & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(B) & \xrightarrow{g} & \operatorname{Spec}(A) \end{array}$$

Let $\mathcal{F}_B \coloneqq g'^*\mathcal{F}$ a quasicoherent $\mathcal{O}_{X_B}$-module. By functoriality there exists an A-module map $H^p(X, \mathcal{F}) \to H^p(X_B, \mathcal{F}_B)$, which induces a B-module homomorphism (a base change homomorphism) $\beta_p : H^p(X, \mathcal{F}) \otimes_A B \to H^p(X_B, \mathcal{F}_B)$. When is this map an isomorphism (i.e. when does cohomology commutes with base change)?

Example. *Let $s \in \operatorname{Spec} A$, $B = k(s)$. Then $X_B = f^{-1}(s) = X_s$ is just the fibre. Let $i : X_s \hookrightarrow X$, so $\mathcal{F}_B = i^*\mathcal{F} = \mathcal{F}(s)$. So the question is when $H^p(X_s, \mathcal{F}(s)) = H^p(X, A) \otimes_A k(s)$?*

Even for $p = 0$, this needn't hold. Consider $p : X = \mathbb{A}^2\backslash\{0\} \to Y = \mathbb{A}^1$ the first coordinate projection, $\mathcal{F} = \mathcal{O}_X$. Then $H^0(X, \mathcal{O}_X) = k[t_1, t_2]$. Let s be the origin in $\mathbb{A}^1$. Then:

$$H^0(X, \mathcal{O}_X) \otimes k(s) = k[t_2]$$

But $X_s = \mathbb{A}^1\backslash\{0\}$, so

$$H^0(X_s, \mathcal{O}_{X_s} = k[t_2, t_2^{-1}])$$

We approach this using Čech cohomology. Consider $(U_i)_{0 \le i \le N}$ be an open affine cover of X, $U_i = \operatorname{Spec} R_i$, $U_I = \cap_{i \in I} U_i = \operatorname{Spec} R_I$ is affine. Then $X_B = \cap U_i'$, where $U_I' = U_I \times_{\operatorname{Spec} A} \operatorname{Spec} B = \operatorname{Spec}(R_I \otimes B)$. So $\mathcal{F}_B(U_I') = \mathcal{F}(U_I) \otimes_A B$. So

$$\check{C}^\bullet((U_i'), \mathcal{F}_B) = \check{C}^\bullet((U_i), \mathcal{F}) \otimes_A B$$

and we have map β_*

$$H^*(X_B, \mathcal{F}_B) = H^*(\check{C}^\bullet((U_i), \mathcal{F}) \otimes_A B) \leftarrow H^*(\check{C}^\bullet((U_i), \mathcal{F})) \otimes_A B = H^*(X, \mathcal{F}) \otimes_A B$$

One important case when β_* is always an isomorphism:

Theorem 4.13 (Flat base change)**.** *If B is a flat A-algebra, then*

$$H^*(X, \mathcal{F}) \otimes_A B \simeq H^*(X_B, \mathcal{F}_B)$$

for any X and any quasicoherent sheaf $\mathcal{F}$.

Proof. By flatness, we know tensoring with B is exact, so it preserves cohomology. □

Example.

1. *If $X \to \operatorname{Spec} k$, K/k any field extension. Then $H^*(X_K, \mathcal{F}_K) = H^*(X, \mathcal{F}) \otimes_k K$*
2. *$B = A_\mathfrak{p}$ localization. $H^*(X, \mathcal{F})_\mathfrak{p} = H^*(X \otimes_A \operatorname{Spec} A_\mathfrak{p}, \mathcal{F}_{A_\mathfrak{p}})$. So we can often reducing understanding cohomology locally to the case of a local base.*

In general, we will replace $\check{C}^\bullet((U_i), \mathcal{F})$ by some smaller complex which computes the cohomology.

Assume f **proper**, $\mathcal{F}$ **coherent and flat** over A. Then $H^p(X, \mathcal{F})$ is then a finite A-module, and vanishes for p sufficiently large (e.g. if $i > N$).

Theorem 4.14. *Assume $H^p(X, \mathcal{F}) = 0$ for all $p > n$. Then there exists a complex $L^0 \to L^1 \to \cdot \to L^n \to 0 \to \cdots$ of finite flat A-module, and for any A-algebra B, we have an isomorphism $H^*(L^\bullet \otimes_A B) \simeq H^*(X_B, \mathcal{F}_B)$, functorial in B.*

Proof. We know that for a finite open affine cover (U_i) of X

$$H^*(X_B, \mathcal{F}_B) = H^*(\check{C}^\bullet((U_i), \mathcal{F}) \otimes_A B)$$

We have the following theorem:

Theorem. *Let A be noetherian, and a complex $K^\bullet$ of A-modules. Assume that $H^p(K^\bullet) = 0$ for all $p > m$ and $H^p(K^\bullet)$ is finite for all p. Then*

1. *there exists a complex $L^0 \to L^1 \to \cdots \to L^m \to 0$ of finite A-modules with $L^1, L^2, \cdots$ free , and a morphism $f : L^\bullet \to K^\bullet$ of complexes such that the induced map on the level of cohomology is an isomorphism.*

2. *Suppose for all $p \geq 0$, K^p is A-flat and zero for p sufficiently large. Then L^0 is flat and for every A-module M,*

   $$H^*(f \otimes 1_M) : H^*(L^\bullet \otimes_A M) \simeq H^*(K^\bullet \otimes_A M)$$

   is an isomorphism.

Proof. 1. Find L^m as follows. Choose L^m to be free and finite, with surjection $\overline{f^m} : L^m \to H^m(K^\bullet)$. Since L^m is free and $\ker d_K^m \to H^m(K^\bullet)$ is surjective, we can lift $\overline{f^m}$ to $f^m : L^m \to \ker d_K^m$.

Then we work down, suppose fthat for some $0 < n \leq m$ we have constructed:

$$\begin{array}{ccccccccc} & & L^n & \longrightarrow & L^{n+1} & \longrightarrow & \cdots & \longrightarrow & L^m \\ & & \downarrow{\scriptstyle f^n} & & \downarrow{\scriptstyle f^{n+1}} & & & & \downarrow{\scriptstyle f^m} \\ \cdots & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & \cdots & \longrightarrow & K^m \longrightarrow K^{m+1} \end{array}$$

such that L^p are finite and free, and $H^p(L^\bullet) \simeq H^p(K^\bullet)$ for $n < p \leq m$, and $\ker d_L^n \twoheadrightarrow H^n(K^\bullet)$ is surjective.

Then we want to construct $d_L^{n-1} : L^{n-1} \to L^n$ and $f^{n-1} : L^{n-1} \to K^{n-1}$.

Choose finite free A-module P with an exat sequence

$$P \xrightarrow{g} \ker d_L^n \to H^n(K) \to 0$$

Then $f^n \circ g(P) \subset \text{im } d_K^{n-1} \subset K^n$. So we can lift g to $\widetilde{g} : P \to K^{n-1}$ such that the diagram

$$\begin{array}{ccc} P & \xrightarrow{g} & L^n \\ \downarrow{\scriptstyle \widetilde{g}} & & \downarrow{\scriptstyle f^n} \\ K^{n-1} & \xrightarrow{d_K^{n-1}} & K^n \end{array}$$

commutes. We choose Q and map $h : Q \to \ker d_K^{n-1} \subset K^{n-1}$ such that

$$Q \to \ker d_K^{n-1} \to H^{n-1}(K^\bullet)$$

is surjective. Let:

$$\begin{array}{ccccc} L^{n-1} = Q \oplus P & \xrightarrow{(0,g)} & L^n & \longrightarrow & L^{n+1} \\ \downarrow{\scriptstyle f^{n-1}=(h,\widetilde{g})} & & \downarrow{\scriptstyle f^n} & & \downarrow{\scriptstyle f^{n+1}} \\ K^{n-1} & \xrightarrow{d_K^{n-1}} & K^n & \longrightarrow & K^{n+1} \end{array}$$

By construction, we have $H^n(f)$ is an isomorphism, and $\ker d_L^{n-1} \twoheadrightarrow H^{n-1}(K^\bullet)$

At the last step, we get finite free $L^0 \xrightarrow{k} H^0(K^\bullet) = \ker d_K^0$. Replace L^0 by a quotient of $L^0/\ker(k)$, hence we're done.

2. Let $C^\bullet$ be the mapping cone:

$$C^p = L^p \oplus K^{p-1} \to L^{p+1} \oplus K^p = C^{p+1}$$

by sending (x, y) to $(dx, f(x) - dy)$. This is a complex. And there exists a short exact sequence of complexes

$$0 \to (K^\bullet[-1], -d_K) \to C^\bullet \to L^\bullet \to 0 \qquad (*)$$

giving a long exact sequence of cohomology

$$\to H^{n-1}(K) \to H^n(C) \to H^n(L) \xrightarrow{\delta} H^n(K) \to$$

and easily see that $\delta^n = H^n(f)$, which is an isomorphism. This forces

$$0 \to C^0 \to C^1 \to \cdots \to C^N \to 0$$

to be exact. But C^p are all flat for all $p > 0$, so $C^0 = L^0$ is flat. Then $(*)$ is exact after tensoring with any A-module M, hence:

$$H^n(C^\bullet \otimes M) \to H^n(L^\bullet \otimes M) \xrightarrow{H^n(f \otimes 1_M)} H^n(K^\bullet \otimes M) \to H^{n+1}(C^\bullet \otimes M)$$

But $C^\bullet \otimes M$ is an exact sequence, this shows $H^n(f \otimes 1_M)$ is an isomorphism.

□

The Theorem above implies the original theorem. □

Remark. *1. In fact for any A-module M, $H^*(L^\bullet \otimes_A M) \simeq H^*(X, \mathcal{F} \otimes_{\mathcal{O}_X} f^*\widetilde{M})$.*

2. We can arrange moreover that L^p is finite and free for all $p > 0$.

First look at H^0. We have

$$H^0(X_B, \mathcal{F}_B) = \ker[L^0 \otimes B \xrightarrow{d_L^0 \otimes 1} L^1 \otimes B]$$

and for any finite flat A-module M, $M \otimes_A B = \mathrm{Hom}_A(M^\vee, A) \otimes_A B = \mathrm{Hom}_A(M^\vee, B)$. Let $Q = \mathrm{coker}\ [(L^1)^\vee \xrightarrow{{}^t d_L^0} (L^0)^\vee]$, we get:

Corollary 4.15. *There exists a finite A-module Q such that for all B,*

$$H^0(X_B, \mathcal{F}_B) = Hom_A(Q, B)$$

functorial in B. So Q represents the functor $B \mapsto H^0(X_B, \mathcal{F}_B)$.

Let $\mathcal{F}(s)$ be the pullback of $\mathcal{F}$ to $X_s = f^{-1}(s)$, for $s \in \mathrm{Spec}\ A$. This is a coherent $\mathcal{O}_{X_s}$-module.

Corollary 4.16 (semicontinuity for H^0)**.** *For every $r \geq 0$, the set*

$$Z_r = \{s \in \mathrm{Spec}\ A | \dim_{k(s)} H^0(X_s, \mathcal{F}(s)) \geq r\}$$

is closed.

Proof. $H^0(X_s, \mathcal{F}(s)) = \mathrm{Hom}_A(Q, k(s))$ for some finite A-module Q.

So it also equals $\mathrm{Hom}_{k(s)}(Q \otimes k(s), k(s))$, hence has dimension $\dim_{k(s)}(Q \otimes k(s))$. So Z_r is closed. □

Remark. *More generally, this holds for all higher cohomology.*

Corollary 4.17. *Assume* Spec *A is connected. Then $\chi(X_s, \mathcal{F}_s)$ is independent of s, for f proper and $\mathcal{F}$ is A-flat. Moreover, if $X \subset \mathbb{P}_A^N$ is projective, then the Hilbert polynomial $P(X_s, \mathcal{F}(s), t)$ is independent of s.*

Proof. The first statement implies the second by definition of Hilbert polynomial (apply the first statement to $\mathcal{F}(n)$). For the first statement, covering Spec A by possibly open affines, may assume L^i are all free.

$$\chi(X_s, \mathcal{F}(s)) = \sum_{p \geq 0} (-1)^p \dim H^p(X_s, \mathcal{F}(s)) = \sum (-1)^p \dim_{k(s)} H^p(L^\bullet \otimes_A k(s))$$

But by homological algebra:

$$\sum (-1)^p \dim_{k(s)} H^p(L^\bullet \otimes_A k(s)) = \sum (-1)^p \dim_{k(s)} L^p \otimes_A k(s)$$

which is independent of s. □

At the top, if $H^p(X, \mathcal{F}) = 0$ for all $p > n$, then for $p > n$, $L^p = 0$, so $H^p(X_B, \mathcal{F}_B) = 0$ for all $p > n$. In particular, $H^p(X_s, \mathcal{F}(s)) = 0$ for all $s, p > n$. Then

$$H^n(X_B, \mathcal{F}_B) = \text{coker } [L^{n-1} \otimes_A B \to L^n \otimes_A B] = \text{coker } [L^{n-1} \to L^n] \otimes_A B = H^n(X, \mathcal{F}) \otimes_A B$$

,since tensor product is right exact. So in degree n, β_* is an isomorphism.

Suppose A is reduced, then

$$A \to \prod_{\mathfrak{p} \in \text{Spec } A} A/\mathfrak{p} = \prod_{s \in \text{Spec } A} k(s)$$

is injective. So as $H^n(X, \mathcal{F})$ is a finite A-module, we know:

$$H^n(X, \mathcal{F}) = 0 \iff \forall s \in \text{Spec } A, H^n(X, \mathcal{F}) \otimes k(s) = 0 \iff \forall s, H^n(X_s, \mathcal{F}(s)) = 0$$

If these equivalent conditions hold, then we can replace n by $n-1$, so by descending induction:

Corollary 4.18. *Assuming A reduced, let $p > 0$. TFAE:*

1. *for any $i \geq p$, $s \in \text{Spec } A$, $H^i(X_s, \mathcal{F}(s)) = 0$.*
2. *for any $i \geq p$, $H^i(X, \mathcal{F}) = 0$.*

5 Group Schemes

Fix a field k, Let Sch/k be the category of k-schemes and k-morphisms. Let Aff/k be the category of k-schemes which are affine. Write $X(S)$ for the S-valued points of X. For $S = \text{Spec } R$, we also write $X(R) = X(S)$.

Example. *Let $X = V(I) \subset \mathbb{A}^N_k$. Then $X(R) = \{(x_1, \cdots, x_N) \in R^N | \forall f \in I, f(\underline{x}) = 0\}$.*

Definition. A k-group scheme is a scheme $G \in \mathsf{Sch}/k$ together with a k-morphism $m : G \times_k G \to G$, such that for every k-algebra R, $m_R : G(R) \times G(R) \to G(R)$ makes $G(R)$ into a group.

Typically, the "topological" points of G doesn't form a group (e.g. $\mathbb{G}_a \simeq \mathbb{A}^1_k$ can't make the points of this into a group in any sensible way).

Functor of Points

For a general scheme X, the point of the underlying space don't determine X. But R-points do:

Let $X \in \mathsf{Sch}/k$, for every $S \in \mathsf{Sch}/k$, we have $X(S) = \text{Mor}_k(S, X)$. If $S' \to S$, we get by composition map $X(S) \to X(S')$, compatible with $S'' \to S' \to S$. Namely, we have a functor

$$h_X : (\mathsf{Sch}/k) \to (\mathsf{Sets})^{\text{op}}$$

and

$$(\mathsf{Aff}/k)^{\text{op}} \cong (k - \mathsf{algebras}) \to (\mathsf{Sets})$$

Lemma 5.1 (Yoneda Lemma)**.**

$$\mathrm{Mor}_k(X,Y) \simeq \mathrm{Nat}(h_X, h_Y) \simeq \{\textit{families of maps } X(S) \to Y(S) \textit{ compatible with } S' \to S\}$$

Lemma 5.2 (Yoneda for schemes)**.** *Same holds for we restrict h_X to affine schemes.*

Proof. Suppose have a compatible family $f_S : X(S) \to Y(S)$ for $S = \mathrm{Spec}\ R$ affines. We need to construct a morphism $f : X \to Y$. Take $X = \bigcup_{U \in \mathcal{U}} U$ open affine cover. For $U \in \mathcal{U}$, take $j_U : S = U \hookrightarrow X$. Then $f_U(j_U) \in Y(U)$. So we have $U \to Y$ for every $U \in \mathcal{U}$. By the functoriality of f_S, $f_U(j_U)$ and $f_{U'}(j'_U)$ agrees with $f_V(j_V)$ for every $V \subset U \cap U'$. So $f_U(j_U)$ glue to give a morphism $X \to Y$. □

Proposition 5.3. *Let G be a k-group scheme. Then:*

1. *For all $S \in \mathsf{Sch}/k$, $G(S)$ is a group.*
2. *For all morphism $S' \to S$, $G(S) \to G(S')$ is a homomorphism of groups.*

Proof. For $S' \to S$ morphism of affine schemes $\mathrm{Spec}\ R' \to \mathrm{Spec}\ R$. Then $G(R) \to G(R')$ is a homomorphism is equivalent to the right hand square in the following diagram commutes:

$$\begin{array}{ccccc} (G\times G)(R) & \xrightarrow{\sim} & G(R)\times G(R) & \longrightarrow & G(R) \\ \downarrow & & \downarrow & & \downarrow \\ (G\times G)(R') & \xrightarrow{\sim} & G(R')\times G(R') & \longrightarrow & G(R') \end{array}$$

The left hand square commutes by definition of product; the outer square commutes trivially. So the right hand square indeed commutes. This proves 2 for affine schemes S', S.

Rest of proof is an exercise. □

Corollary 5.4. *There exists $e \in G(k)$ and $i : G \to G$ such that for all S, $e \mapsto 1_{G(S)}$, and $i_S : G(S) \to G(S)$ is inverse.*

Proof. Let $e = 1_{G(k)}$. Since $G(k) \to G(S)$ is a homomorphism, so takes e to identity in $G(S)$. The second statement is just Yoneda: take i to be the inverse of id_G in the group $G(G)$. Namely $\mathrm{id}_G \times i \mapsto 1_{G(G)}$ in $G(G)$. Now we fix $g \in G(S)$, this induces a homomorphism of groups

$$\varphi : G(G) \to G(S), \lambda \mapsto \lambda \circ g$$

Hence we know:

$$g \times (i \circ g) = g \times i_S(g) = 1_{G(S)}$$

as required. □

Definition. (Alternative for Group Scheme) A group scheme is a system

$$(G, m : G \to G, e \in G(k), i : G \to G)$$

satisfying suitable axioms (for groups). This is a group object in the category of Sch/k.

Definition. (Alternative for Group Scheme) A group scheme is a representable functor $(\mathsf{Sch}/k)^{\mathrm{op}} \to \mathsf{Groups}$

Definition. A homomorphism of group scheme is a morphism $f : G \to G'$ such that for all $R \in k - \mathsf{algebras}$, $f_R : G(R) \to G'(R)$ is a group homomorphism. Equivalently, $f_S : G(S) \to G'(S)$ is a homomorphism for $S \in \mathsf{Sch}/k$. Equivalently,

$$\begin{array}{ccc} G \times G & \xrightarrow{f \times f} & G' \times G' \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle m'} \\ G & \xrightarrow[f]{} & G' \end{array}$$

commutes.

Definition. A closed subgroup scheme of k-group scheme G is a closed subscheme $i : H \hookrightarrow G$ such that for every R, $H(R) \subset G(R)$ is a subgroup. If so, H is a group scheme, and i is a homomorphism of group schemes.

Example. *1.* $e : \operatorname{Spec} k \hookrightarrow G$ *is a closed subgroup scheme.*

2. *Let* $f : G \to G'$ *be any homomorphism. Define* $\ker(f) = G \times_{G'} e'$*. By definition of fibre product, for all* S, $\ker(f)(S) = \ker(f_S : G(S) \to G(S'))$*. Since* e' *is a closed immersion,* $\ker f$ *is a closed subscheme. So* $\ker f$ *is a closed subgroup scheme.*

3. *For every* R*, we have* $\mathrm{GL}_n(R) \to R^\times = \mathbb{G}_R$ *is functorial in* R*. So by Yoneda, we have a homomorphism of group schemes* $\det : \mathrm{GL}_n \to \mathbb{G}_m$*.* $\mathrm{SL}_n := \ker(\det) = \operatorname{Spec} k[x_{ij}]/(\det(t_{ij}) - 1)$

Remark. *Quotients are harder.*

Remark. *Fix a scheme* Y*. A group scheme over* Y *is a* Y*-scheme* $G \to Y$ *and a morphism* $m : G \times_Y G \to G$ *such that for any* Y*-scheme* S*,* $G(S)$ *is a group. Then* $i : G \to G$*,* $e \in G(Y)$*. It is also enough to know* $G(S)$ *is a group for* S *affine.*

Let $x \in G(k)$. The *left translation* morphism $T_x : G \to G$ is the composite

$$G = \operatorname{Spec} k \times_k G \xrightarrow{x \times 1_G} G \times G \xrightarrow{m} G$$

It is the uniqe morphism such that for all $g \in G(S), T_x(g) = m(x, g) = xg$. Obviously, $T_e = 1_G, T_{xy} = T_x \circ T_y$. So T_x is a k-automorphism of G. More generally, we have:

Definition (Translation)**.** Let S be a k-scheme, $x \in G(S)$. Then $T_x : G \times_k S \to G$ is the composite

$$G \times_k S \xrightarrow{1_G \times x} G \times G \xrightarrow{{}^t m} G$$

where ${}^t m : G \times G \to G$ is given on point s by $(g, h) \mapsto hg$. Note this is as before for $S = \operatorname{Spec} k$.

The product $T_{x/S} : G \times S \xrightarrow{(T_x, \mathrm{pr}_2)} G \times S$ satisfies $T_{e/S} = 1_{G\times S}, T_{x/S} \circ T_{y/S} = T_{xy/S}$. So $T_{x/S}$ is an S-automorphism of $G \times_k S$.

Similarly, define the right translation (and equals left translation if G is commutative).

Definition. We call G commutative, if $G(R)$ is commutative for all $R \in k$-algebras.

Remark. *This is equivalent to $m = {}^t m$ by Yoneda.*

If G is a k-group scheme. Let k'/k be any extension. We get $G \times_k \operatorname{Spec} k' = G_{k'}$ is a k'-scheme, with the multiplication:

$$G_{k'} \times_{k'} G_{k'} = (G \times_k G) \times_k k' \to G \times_k k' = G_{k'}$$

Recall: a k-variety is a separated k-scheme of finite type which is geometrically integral.

Definition. A k-group variety (or algebraic group) is a k-group scheme which is a k-variety. An *abelian variety* (AV) over k is a k-group variety which is *proper* over k.

Example. $\mathbb{G}_a, \mathbb{G}_m, \mathrm{GL}_n$ *are group varieties. Simplest nontrivial example of abelian variety is an elliptic curve over k (nonsingular plane cubic with a distinguished k-point). Recall the "Chord and Tangent" construction is a morphism $E \times E \to E$.*

Note product of two group varieties (resp. abelian varieties) is one as well. In particular, there are abelian varieties of any dimension.

Remark. *Some people define algebraic group to be k-group scheme of finite type.*

Proposition 5.5. *Let G be a group variety over k. Then G is smooth over k.*

Proof. Recall enough to check $G_{\overline{k}}$ is smooth over $\overline{k}$. So we may assume $k = \overline{k}$. Then closed points are same as k-points. The set of smooth points is an open dense (so contain a closed point, as G is finite type over k).

If $x \in G(k)$, then $T_x G \simeq G$ takes e to x. So G is smooth at $x \iff G$ is smooth at e. So G is smooth at all closed points, hence G is smooth. □

So $\Omega_{G/k}$ is locally free if G is a k-group variety. In fact $\Omega_{G/k}$ is free for any k-group scheme.

Theorem 5.6 (Mumford's Rigidity Lemma, classical version)**.** *Let X, Y, Z be k-varieties, with X proper. Assume $X(k) \neq \emptyset$. Suppose we have a k-morphism*

$$f : X \times_k Y \to Z$$

Suppose for some $y_0 \in Y$ and $z_0 \in Z$

$$X \times \{y_0\} \subset f^{-1}(z_0)$$

So f collapses $X \times \{y_0\}$ to a point. Then there exists $g : Y \to Z$ such that $f = g \circ \mathrm{pr}_2$. In particular, for all $y \in Y$, $f(X \times \{y\})$ is a single point.

In words: given a family of morphisms $X \to Z$, X proper. If one member of the family is a constant morphism, then so is every member.

Example. *This is an reality check. Let $X = Y = Z = \mathbb{A}^1_k$, $f(x, y) = xy$. Let y be the origin in $Y = \mathbb{A}_1$, then X maps to a single point $0 \in Z = \mathbb{A}_1$. But it doesn't collapse elsewhere.*

Corollary 5.7. *Let X be an abelian variety, G group variety. If $f : X \to G$ is any morphism, $g = f(e) \in G(k)$. Then $T_{g^{-1}} \circ f$ is a homomorphism.*

Proof. Replace f by $T_{g^{-1}} \circ f$, we reduce to showing: if $f(e_X) = e_G \implies f$ is a homomorphism. Consider $h : X \times X \to G$ given by $(x, y) \mapsto f(x)f(y)f(xy)^{-1}$. Note that $h(X \times \{e_X\}) = h(\{e_X\} \times X) = e_G$. Then by the rigidity lemma, $h \equiv e_G$, f is a group homomorphism. □

In particular, if $f : X \to X$ is an isomorphism of schemes taking $e \to e$ (X an abelian variety), then f is an isomorphism of group scheme. So abelian varieties have only one possible group structure, once we fixed an identity element.

Example. *This is a reality check. Let $G = \mathbb{G}_a \times \mathbb{G}_a \times \mathbb{G}_a$, and $G' = \begin{pmatrix} 1 & * & * \\ & 1 & * \\ & & 1 \end{pmatrix}$. They both isomorphic to $\mathbb{A}^3$.*

Corollary 5.8. *An abelian variety is commutative.*

Proof. Consider $i : X \to X$ the inverse map. As $i(e) = e$, this is a homomorphism. Then each $X(R)$ is commutative. □

Lemma 5.9. *Let X be proper k-variety, Y any k-scheme. Then*

$$\mathrm{pr}_{2*}\mathcal{O}_{X\times Y} = \mathcal{O}_Y$$

Proof.

$$\begin{array}{ccc} X \times_k Y & \longrightarrow & X \\ {\scriptstyle \mathrm{pr}_2}\downarrow & & \downarrow{\scriptstyle a_X} \\ Y & \xrightarrow[a_Y]{} & k \end{array}$$

Since Y is trivially flat over Spec k, by flat base change, $\mathrm{pr}_{2*}\mathcal{O}_{X\times Y} = a_Y^* a_{X*}\mathcal{O}_X = \mathcal{O}_Y \otimes_k \Gamma(X, \mathcal{O}_X) = \mathcal{O}_Y$, as X is a proper variety. □

So in the setting of 5.6, $\mathrm{pr}_{2*}\mathcal{O}_{X\times Y} = \mathcal{O}_Y$, and as X is proper, pr_2 is closed. So $p = \mathrm{pr}_2 : X \times Y \to Y$ satisfies the conditions of the following theorem:

Theorem (Rigidity)**.** *Let $p : X \to Y$ be a morphism of schemes with a section $s : Y \to X$. Assume p closed, $p_*\mathcal{O}_X = \mathcal{O}_Y$, and X integral. Let $f : X \to Z$ to some separated scheme Z. Suppose there exists $y \in Y, z \in Z$ (not necessarily closed), such that $X_y = p^{-1}(y) \subset f^{-1}(z)$, then $f = g \circ p$ for some unique $g : Y \to Z$.*

Proof. There are two reductions:

1. Suppose there exists $g : Y \to Z$ with $f = g \circ p$, then $g = g \circ (p \circ s) = f \circ s$. So g is unique, and exists if $f = (f \circ s) \circ p$.

2. Suppose there exists some open neighborhood, $Y' \subset Y$ of y such that $f'|_{X'}$, $X' = p^{-1}(Y')$, factors through some $Y' \to Z$. Then by the first part, we know $f|_{X'} = (f \circ s \circ p)|_{X'}$. Note $X' \neq \emptyset$ (contains $s(y)$), and is dense (X integral). This implies $f = (f \circ s \circ p)$, because X is reduced and Z is separated. [first exercise on separated schemes in Hartshorne]

Now f maps X_y to $\{z\} \subset Z$. Let $W \subset Z$ be an open affine containing z. Then $f^{-1}(W)$ is an open neighborhood of X_y. But since p is closed, this means that $f^{-1}(W) \subset p^{-1}(Y') = X'$ for some open neighborhood Y' of y. [In fact, take $T = p(X \backslash f^{-1}(W)) \subset Y$ is closed and doesn't contain y. Take $Y' = Y \backslash T$.]

Now $f : X' \to W = \operatorname{Spec} B$ say, so given by a homomorphism $B \to \Gamma(X', \mathcal{O}_{X'}) = \Gamma(Y', \mathcal{O}_{Y'})$. So $f|_{X'}$ factors through Y'. We're done by (2). □

6 Seesaw, Cube, and Square

Theorem 6.1 ("Seesaw")**.** *Let X, Y be k-varieties, X proper. Let $\mathcal{L}$ be a line bundle on $X \times_k Y$. Suppose for all closed $y \in Y$, the line bundle $\mathcal{L}(y) = i_y^*\mathcal{L}$ is trivial, where $i_y : X \times \operatorname{Spec} k \xrightarrow{1_X \times y} X \times Y$. Then there exists a line bundle $\mathcal{M}$ on Y such that $\mathcal{L} = \operatorname{pr}_2^*\mathcal{M}$. Moreover, $\mathcal{M} \cong \operatorname{pr}_{2*}\mathcal{L}$ is unique up to isomoprhism.*

Proof. We first show that for all $y \in Y$ has an affine neighborhood V such that $\mathcal{L}|_{X \times_k V} \cong \mathcal{O}_{X \times_k V}$. For this, may assume $Y = \operatorname{Spec} A$ affine. Then there exists finite A-module Q such that $H^0(X \times y, \mathcal{L}(y)) = \operatorname{Hom}_A(Q, k(y)) = (Q \otimes_A k(y))^\vee$. So by assumption, for all maximal $\mathfrak{m} \subset A$, $\dim_{A/\mathfrak{m}}(Q/\mathfrak{m}Q) = 1$. So Q is locally free of rank 1 (because $H^0(\mathcal{L}(y)) = k$ for all closed y, X is proper). So every $y \in Y$ does have an open neighborhood $V = \operatorname{Spec} B$ affine, for which the corresponding B-module Q is free, say generated by u. Then $H^0(X \times V, \mathcal{L}) = \operatorname{Hom}_B(Q, B) = Bu^\vee$. And at each closed $y \in Y$, $u^\vee$ maps to a nonzero section of $H^0(X \times y, \mathcal{L}(y)) = \operatorname{Hom}(\mathcal{O}_{X \times y}, \mathcal{L}(y))$. As $\mathcal{L}(y) = \mathcal{O}_{X \times y}$, $u^\vee$ maps to a nowhere vanishing section, i.e. $u^\vee : \mathcal{O}_{X \times V} \simeq \mathcal{L}|_{X \times V}$. By 5.9, $\operatorname{pr}_{2*}\mathcal{O}_{X \times V} = \mathcal{O}_V$, so the adjunction $\operatorname{pr}_2^*\operatorname{pr}_{2*} : \mathcal{O}_{X \times V} \to \mathcal{O}_{X \times V}$ is an isomorphism, i.e. $\operatorname{pr}_2^*\operatorname{pr}_{2*} : \mathcal{L}|_{X \times V} \simeq \mathcal{L}|_{X \times V}$. For general Y, this shows that $\operatorname{pr}_{2*}\mathcal{L} = \mathcal{M}$ is invertible, and $\operatorname{pr}_2^*\mathcal{M} \cong \mathcal{L}$. If $\mathcal{L} \cong \operatorname{pr}_2^*\mathcal{M}'$ for some $\mathcal{M}' \in \operatorname{Pic}(Y)$, then $\mathcal{M}' \cong \operatorname{pr}_{2*}\mathcal{L}$, i.e. $\mathcal{M} \cong \mathcal{M}'$. □

Corollary 6.2. *Under same hypothesis, if in addition there exists $x \in X(k)$ and $\mathcal{L}|_{x \times Y} \simeq \mathcal{O}_Y$, then $\mathcal{L}$ is trivial.*

Proof. Let $\mathcal{L} = \operatorname{pr}_2^*\mathcal{M}$, and note $\operatorname{pr}_2 \circ (x \times 1_Y) = 1_Y$. So

$$\mathcal{O}_Y = \mathcal{L}|_{x \times Y} = (x \times 1_Y)^*\mathcal{L} = (x \times 1_Y)^*\operatorname{pr}_2^*\mathcal{M} = \mathcal{M}$$

So $\mathcal{L} \cong \operatorname{pr}_2^*\mathcal{O}_Y = \mathcal{O}_{X \times Y}$. □

We'll see in the exercise sheet that we can have proper k-varieties X, Y and $\mathcal{L} \in \operatorname{Pic}(X \times Y)$ such that $\mathcal{L}|_{X \times y}, \mathcal{L}|_{x \times Y}$ are trivial, but $\mathcal{L}$ is nontrivial. In particular, $\operatorname{Pic}(X \times Y) \neq \operatorname{Pic}X \oplus \operatorname{Pic}Y$ in general.

Theorem 6.3 (Theorem of the Cube)**.** *Let X, Y, Z be k-varieties, with X, Y proper. Let $x \in X(k), y \in Y(k), z \in Z(k)$, $\mathcal{L}$ line bundle on $X \times Y \times Z$. Assume each of the line bundles $\mathcal{L}|_{x \times Y \times Z}, \mathcal{L}|_{X \times y \times Z}, \mathcal{L}|_{X \times Y \times z}$ are trivial, then $\mathcal{L}$ is trivial.*

Proof. We will assume $k = \overline{k}$. To prove this, consider other k-schemes Z (typical reduction technique).

1. $Z = \text{Spec } A$ for A finite local k-algebra, so Z is a possibly non-reduced point. We prove this on $\dim_k A \geq 1$. If $\dim A = 1$, so $Z = \text{Spec } k \implies X \times Y \times Z \simeq X \times Y \times z$. There's nothing to prove. Otherwise, choose an ideal $I \subset A$ with $\dim_k I = 1, I = k \cdot t \subset A$. Now look at $Z_1 = \text{Spec } A/I$. We now need the following

   Lemma. *Let V be a proper variety. We have an exact sequence functorial in V:*

   $$0 \to H^1(V, \mathcal{O}_V) \to \text{Pic}(V \times Z) \to \text{Pic}(V \times Z_1)$$

   Proof. (of lemma) Note $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ for any scheme X. Note $I^2 = 0 \implies (1+a)(1+b) = 1 + (a+b)$ for all $a, b \in I$, hence we have an exact sequence of abelian groups

   $$0 \to I \to A^* \to (A/I)^* \to 0$$

   where the first map is $a \mapsto 1 + a$, and

   $$0 \to I\mathcal{O}_{V \times Z} \to \mathcal{O}_{V \times Z}^* \to \mathcal{O}_{V \times Z_1}^* \to 0$$

   is an exact sequence of sheaves of abelian groups on $V \times Z \simeq V$. Now $\mathcal{O}_V \simeq I\mathcal{O}_{V \times Z}$ via multiplication by t. Applying the long exact sequence of cohomology:

   $$0 \to H^0(V, \mathcal{O}_V) \to H^0(V \times Z, \mathcal{O}_{V \times Z})^* \to H^0(V \times Z_1, \mathcal{O}_{V \times Z_1})^*$$

   We know:

   $$H^0(V, \mathcal{O}_V) = k$$
   $$H^0(V \times Z, \mathcal{O}_{V \times Z})^* = (H^0(V, \mathcal{O}_V) \otimes A)^* = A^*$$
   $$H^0(V \times Z_1, \mathcal{O}_{V \times Z_1})^* = (H^0(V, \mathcal{O}_V) \otimes (A/I))^* = (A/I)^*$$

   So the last map is surjective. Hence we have

   $$0 \to H^1(V, \mathcal{O}_V) \to \text{Pic}(V \times Z) \to \text{Pic}(V \times Z_1)$$

   (clearly construction functorial in V) □

   By induction, we may assume $\mathcal{L}_{X \times Y \times Z_1}$ trivial. So by the lemma, we have:

$$\begin{array}{ccccccc}
0 & \longrightarrow & H^1(X \times Y, \mathcal{O}) & \longrightarrow & \text{Pic}(X \times Y \times Z) & \xrightarrow{c} & \text{Pic}(X \times Y \times Z_1) \\
 & & \downarrow a & & \downarrow b & & \downarrow \\
0 & \longrightarrow & H^1(X, \mathcal{O}) \oplus H^1(Y, \mathcal{O}) & \longrightarrow & \begin{array}{c}\text{Pic}(X \times Z) \\ \oplus \text{Pic}(Y \times Z)\end{array} & \longrightarrow & \begin{array}{c}\text{Pic}(X \times Z_1) \\ \oplus \text{Pic}(Y \times Z_1)\end{array}
\end{array}$$

where the vertical maps are (y^*, x^*). By Künneth formula, plus $H^0(X, \mathcal{O}) = H^0(Y, \mathcal{O}) = k$, gives an isomoprhism

$$H^1(X, \mathcal{O}) \oplus H^1(Y, \mathcal{O}) \simeq H^1(X \times Y, \mathcal{O})$$

since sum of pr_1^* and pr_2^* is an inverse to a. As $\mathcal{L} \in \mathrm{Pic}(X \times Y \times Z), b(\mathcal{L}) = c(\mathcal{L}) = 0$, this shows $\mathcal{L}$ is trivial by diagram chase.

2. $Z = \mathrm{Spec}\ A$ for A Noetherian local k-algebra. Let $Z_n = \mathrm{Spec}\ A/\mathfrak{m}_A^n$. Then $A/\mathfrak{m}_A^n$ is finite. So by the previous step, we know $\mathcal{L}|_{X \times Y \times Z_n}$ is trivial for any n. From base change, there exists A-modules Q, Q' such that for any $A \to B$, $H^0(\mathcal{L}_B) = \mathrm{Hom}_A(Q, B), H^0(\mathcal{L}_B^\vee) = \mathrm{Hom}_A(Q', B)$. As $\mathcal{L}|_{X \times Y \times z} \simeq \mathcal{O}$, we know Q, Q' are both cyclic modules.

   Since $\mathcal{L}|_{X \times Y \times Z_n}$ is trivial, $Q \otimes A/\mathfrak{m}^n \simeq A/\mathfrak{m}^n$. So $\mathrm{ann}_A(Q) \subset \bigcap_n \mathfrak{m}_A^n = \{0\}$. So $Q \simeq A \simeq Q'$. Hence by seesaw, $\mathcal{L} \simeq \mathcal{O}_{X \times Y \times Z}$.

3. Z is arbitrary variety. Then the previous part tells us $\mathcal{L}|_{X \times Y \times \mathrm{Spec}\ \mathcal{O}_{Z,z}}$ is trivial. Note the generic point $\eta \in \mathrm{Spec}\ \mathcal{O}_{Z,z}$, we see $\mathcal{L}_{X \times Y \times z'}$ is trivial for all $z' \in Z$, as the set of such z' is closed. As $\mathcal{L}|_{x \times y \times Z}$ is trivial, we can then use seesaw to see that $\mathcal{L}$ is trivial.

□

Remark. *(of the lemma) Take* $Z = \mathrm{Spec}\ k[\varepsilon]/(\varepsilon^2), Z_1 = \mathrm{Spec}\ k$. *We get*

$$0 \to H^1(V, \mathcal{O}_V) \to \mathrm{Pic}(V \times \mathrm{Spec}\ k[\varepsilon]) \to \mathrm{Pic}V \to 0$$

Note the last map is clearly surjective. This says that the tangent space to Pic *is* $H^1(V, \mathcal{O}_V)$. *In fact there is a group scheme* $\mathrm{Pic}_{V/k}$ *with* $\mathrm{Pic}_{V/k}(k) = \mathrm{Pic}(V)$, *and* $\mathrm{Pic}_{V/k}(k[\varepsilon]) = \mathrm{Pic}(V \times \mathrm{Spec}\ k[\varepsilon]) = T_{\mathrm{Pic}_{V/k}, 0}$

Remark. *In fact, Z can be a more general k-scheme not reduced (needed for proof). Seesaw theorem also holds for more general Y.*

Corollary 6.4. *Let X be an abelian variety over k, Y any k-scheme. Given $f, g, h : Y \to X$, $\mathcal{L}$ line bundle on X. Then*

$$\mathcal{M} = \mathcal{M}_{f,g,h} \coloneqq (f+g+h)^*\mathcal{L} \otimes (f+g)^*\mathcal{L}^\vee \otimes (f+h)^*\mathcal{L}^\vee \otimes (g+h)^*\mathcal{L}^\vee \otimes f^*\mathcal{L} \otimes g^*\mathcal{L} \otimes h^*\mathcal{L}$$

is trivial.

Proof. Consider $Y = X \times X \times X$, and f, g, h are just three projections $(\mathrm{pr}_1^3, \mathrm{pr}_2^3, \mathrm{pr}_3^3)$.

$$\mathcal{M}|_{X \times X \times e} = q^*(\mathcal{M})$$

$$q : X \times X \to X \times X \times X, (x, y) \mapsto (x, y, e)$$

$$(\mathrm{pr}_1^3 + \mathrm{pr}_2^3 + \mathrm{pr}_3^3) \circ q = (\mathrm{pr}_1^3 + \mathrm{pr}_2^3) \circ q = m : (x, y) \mapsto x + y$$

$$(\mathrm{pr}_1^3 + \mathrm{pr}_3^3) \circ q = \mathrm{pr}_1^3 \circ q = \mathrm{pr}_1^2 : (x, y) \mapsto x$$

$$(\mathrm{pr}_2^3 + \mathrm{pr}_3^3) \circ q = \mathrm{pr}_2^3 \circ q = \mathrm{pr}_2^2 : (x, y) \mapsto y$$

$$\mathrm{pr}_3^3 \circ q : (x, y) \mapsto e$$

So $\mathcal{M}|_{X\times X\times e} = m^*\mathcal{L} \otimes m^*\mathcal{L}^\vee \otimes \mathrm{pr}_1^{2*}\mathcal{L}^\vee \otimes \mathrm{pr}_2^{2*}\mathcal{L}^\vee \otimes \mathrm{pr}_1^{2*}\mathcal{L} \otimes \mathrm{pr}_2^{2*}\mathcal{L} \otimes \mathcal{O} = \mathcal{O}$. By symmetry, we know $\mathcal{M}|_{X\times e\times X} \simeq \mathcal{M}|_{e\times X\times X} \simeq \mathcal{O}_{X\times X}$. So by Cube, $\mathcal{M}$ is trivial.

For general, $f, g, h : Y \to X$, $\mathcal{M}_{f,g,h} = (f, g, h)^*\mathcal{M}_{\mathrm{pr}_1,\mathrm{pr}_2,\mathrm{pr}_3}$, $(f, g, h) : Y \to X \times X \times X$, so is trivial. □

Theorem 6.5 (Theorem of the square)**.** *Let X be an abelian variety. $x, y \in X(k), \mathcal{L} \in \mathrm{Pic}(X)$. Then $T_{x+y}^*\mathcal{L} \simeq T_x^*\mathcal{L} \otimes T_y^*\mathcal{L} \otimes \mathcal{L}^\vee$*

Proof. Let $Y = X$, $f = x : X \to \mathrm{Spec}\ k \xrightarrow{x} X$, $g = y : X \to \mathrm{Spec}\ k \xrightarrow{y} X$, $h = 1_X$. Then:

$$f + h = T_x, g + h = T_y, f + g + h = T_{x+y}, f + g = \text{constant morphism } x + y$$

Note if $p : X \to X$ is constant, $p^*\mathcal{L} \simeq \mathcal{O}_X$. Then:

$$\mathcal{M}_{f,g,h} = T_{x+y}^*\mathcal{L} \otimes T_x^*\mathcal{L}^\vee \otimes T_y^*\mathcal{L}^\vee \otimes \mathcal{O}_X^\vee \otimes \mathcal{O}_X \otimes \mathcal{O}_X \otimes \mathcal{L}.$$

As this is trivial, we get the desired equality. □

Corollary 6.6. *Let X be an abelian variety, $n \in \mathbb{Z}$. $[n] : X \to X$ multiplication by n morphism. Let $\mathcal{L} \in \mathrm{Pic}(X)$. Then $[n]^*\mathcal{L} = \mathcal{L}^{\otimes \frac{n(n+1)}{2}} \otimes (i^*\mathcal{L})^{\otimes \frac{n(n-1)}{2}}$, where $i = [-1] : X \to X$. In particular, if $\mathcal{L} = i^*\mathcal{L}$ (i.e. $\mathcal{L}$ is symmetric), then $[n]^*\mathcal{L} = \mathcal{L}^{\otimes n^2}$; if $\mathcal{L} = i^*\mathcal{L}^\vee$ (i.e. $\mathcal{L}$ is antisymmetric), then $[n]^*\mathcal{L} = \mathcal{L}^{\otimes n}$.*

Proof. We prove by induction. For $n = 0$, there's nothing to prove. For $n = 1$, it's trivial. Apply Corollary of Cube with $f = [n-1], g = [1] = 1_X, h = i = [-1] : X \to X$:

$$[n-1]^*\mathcal{L} \simeq [n]^*\mathcal{L} \otimes [n-2]^*\mathcal{L} \otimes \mathcal{O}_X \otimes [n-1]^*\mathcal{L}^\vee \otimes \mathcal{L}^\vee \otimes (i^*\mathcal{L})^\vee$$

So:

$$[n]^*\mathcal{L} \simeq [n-1]^*\mathcal{L}^{\otimes 2} \otimes [n-2]^*\mathcal{L}^\vee \otimes \mathcal{L} \otimes i^*\mathcal{L}$$

We conclude by computation with the induction hypothesis. This shows for $n \geq 0$. For the case $n < 0$, use $[n] = [-1][-n]$. □

7 Pic of an abelian variety

$\mathrm{Pic}(X)$ is an abelian group, with inverse $\mathcal{L} \mapsto \mathcal{L}^\vee$, unit $\mathcal{O}_X$. Contravariant: $f : X \to Y \implies f^* : \mathrm{Pic}(Y) \to \mathrm{Pic}(X)$ is a homomorphism.

It is convenient to work over $\overline{k}$ sometimes. This loses no information.

Proposition 7.1. *Let X be a k-variety. Then $\mathrm{Pic}(X) \hookrightarrow \mathrm{Pic}(X_{\overline{k}})$ is injective.*

Proof. Let $\mathcal{L} \in \mathrm{Pic}(X)$, $\mathcal{L}_{\overline{k}} \simeq \mathcal{O}_{X_{\overline{k}}}$. It suffices to show $\mathcal{L} \simeq \mathcal{O}_X$. But $\mathcal{L} \simeq \mathcal{O}_X \iff H^0(X, \mathcal{L}) \neq 0 \neq H^0(X, \mathcal{L}^\vee) \iff H^0(X_{\overline{k}}, \mathcal{L}_{\overline{k}}) \neq 0 \neq H^0(X_{\overline{k}}, \mathcal{L}_{\overline{k}}^\vee)$ by flat base change. So $\iff \mathcal{L}_{\overline{k}} = \mathcal{O}_{\overline{k}}$. □

From now on, except when stated otherwise, assume $k = \overline{k}$.

Next, we want to use the theorem of square (and more) to show $X(k) \twoheadrightarrow \mathrm{Pic}^0(X) \subset \mathrm{Pic}(X)$ with finite kernel.

Proposition 7.2. *Let X be an abelian variety, $\mathcal{L}$ line bundle on X.*

1. *For $x \in X(k)$, set $\varphi_{\mathcal{L}}(x) = T_x^*\mathcal{L} \otimes \mathcal{L}^\vee$. Then $\varphi_{\mathcal{L}} : X(k) \to \mathrm{Pic}(X)$ is a homomorphism.*
2. *Let*

$$K(\mathcal{L}) \coloneqq \ker \varphi_{\mathcal{L}} \subset X(k)$$

$$\mathrm{Pic}^0(X) \coloneqq \{\mathcal{L} \in \mathrm{Pic}X | \varphi_{\mathcal{L}} = 0\} = \{\mathcal{L} \in \mathrm{Pic}(X) \mid \forall x \in X(k), T_x^*\mathcal{L} \simeq \mathcal{L}\}$$

is a subgroup, and define the Néron-Sereri group of X to be

$$\mathrm{NS}(X) \coloneqq \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$$

As $k = \overline{k}$, $X(k)$ is the collection of closed points of X.

In general, define $\varphi_{\mathcal{L}} : X(\overline{k}) \to \mathrm{Pic}X_{\overline{k}}$ the same way.

Proof. 1. $\varphi_{\mathcal{L}}(x+y) = T_{x+y}^*\mathcal{L} \otimes \mathcal{L}^\vee = T_x^*\mathcal{L} \otimes T_y^*\mathcal{L} \otimes \mathcal{L}^\vee \otimes \mathcal{L}^\vee = \varphi_{\mathcal{L}}(x) + \varphi_{\mathcal{L}}(y)$ by the theorem of square.

2. Note $\mathrm{Pic}(X) \to \mathrm{Pic}(X), \mathcal{L} \to T_x^*\mathcal{L}$ is a group homomorphism. So

$$\mathrm{Pic}(X) \to \mathrm{Hom}(X(k), \mathrm{Pic}(X))$$
$$\mathcal{L} \to \varphi_{\mathcal{L}}$$

is a group homomorphism, with kernel $\mathrm{Pic}^0(X)$.

□

Definition. Let $\mathcal{L} \in \mathrm{Pic}(X)$. Define $\Lambda(\mathcal{L}) = m^*\mathcal{L} \otimes \mathrm{pr}_1^*\mathcal{L}^\vee \otimes \mathrm{pr}_2^*\mathcal{L}^\vee$ (Mumford line bundle) on $X \times_k X$, for m multiplication.

Proposition 7.3. 1. *For all $x \in X(k)$, $\Lambda(\mathcal{L})|_{X\times x} = T_x^*\mathcal{L} \otimes \mathcal{L}^\vee = \Lambda|_{x\times X}$, and $K(\mathcal{L}) = \{x \in X(k) \mid \Lambda(\mathcal{L})|_{X\times x} \cong \mathcal{O}_X\}$*

2. *$\mathcal{L} \in \mathrm{Pic}^0(X) \iff \Lambda(\mathcal{L}) \cong \mathcal{O}_{X\times X}$*

Proof. 1. Let $x \in X(k)$, consider $(1_X, x) : X \to X \times X$. Then $m \circ (1_X, x) = T_x : X \to X$; $\mathrm{pr}_1 \circ (1_X, x) = 1_X$; $\mathrm{pr}_2 \circ (1_X, x) =$constant morphism x. Hence:

$$\Lambda(\mathcal{L})|_{X\times x} = (1_X, x)^*\Lambda(\mathcal{L}) = T_x^*\mathcal{L} \otimes \mathcal{L}^\vee = \Lambda(\mathcal{L})|_{x\times X}.$$

by symmetry (X is commutative).

2. $\Lambda(\mathcal{L})|_{e\times X} = \mathcal{O}_X$ as $T_e = 1_X$. So by Seesaw,

$$\Lambda(\mathcal{L}) \simeq \mathcal{O}_{X\times X} \iff \forall x, \Lambda(\mathcal{L})|_{X\times x} \simeq \mathcal{O}_X \iff \forall x, \varphi_{\mathcal{L}}(x) = 0 \iff \mathcal{L} \in \mathrm{Pic}^0(X)$$

□

Proposition 7.4. *1. For all $\mathcal{L}$,* im $(\varphi_{\mathcal{L}}) \in \mathrm{Pic}^0(X)$

2. If $\mathcal{L} \in \mathrm{Pic}^0(X)$, then $i^*\mathcal{L} \simeq \mathcal{L}^\vee$, where $i = [-1]$.

Proof. 1. Let $\mathcal{M} = \varphi_{\mathcal{L}}(x) = T_x^*\mathcal{L} \otimes \mathcal{L}^\vee$, $\mathcal{L} \in \mathrm{Pic}(X), x \in X(k)$. Then for all $y \in X(k)$,

$$\varphi_{\mathcal{M}}(y) = T_y^*(T_x^*\mathcal{L} \otimes \mathcal{L}^\vee) \otimes (T_x^*\mathcal{L}^\vee \otimes \mathcal{L}) = T_{x+y}^*\mathcal{L} \otimes T_x^*\mathcal{L}^\vee \otimes T_y^*\mathcal{L}^\vee \otimes \mathcal{L} \simeq \mathcal{O}_X$$

by Theorem of Square. So $\mathcal{M} \in \mathrm{Pic}^0(X)$.

2. Let $\mathcal{L} \in \mathrm{Pic}^0(X)$, we know $m^*\mathcal{L} = \mathrm{pr}_1^*\mathcal{L} \otimes \mathrm{pr}_2^*\mathcal{L}$. Consider the codifference map:

$$d : X \to X \times X, x \mapsto (x, -x)$$

$d^*m^*\mathcal{L} = \mathcal{O}_X$, as $m \circ d$ is the constant morphism $e : X \to X$.
So $\mathcal{O}_X = d^*(\mathrm{pr}_1^*\mathcal{L} \otimes \mathrm{pr}_2^*\mathcal{L}) = \mathcal{L} \otimes i^*\mathcal{L}$. as $d \circ \mathrm{pr}_1 = 1_X$, $d \circ \mathrm{pr}_2 = i$.

□

Aim

Theorem. *If $\mathcal{L}$ is ample, then $K(\mathcal{L})$ is finite, and we have a short exact sequence*

$$0 \to K(\mathcal{L}) \to X(k) \xrightarrow{\varphi_{\mathcal{L}}} \mathrm{Pic}^0(X) \to 0$$

Exercise. Work this out for elliptic curves.

Proposition 7.5. *Let $\mathcal{L} \in \mathrm{Pic}(X)$, $K(\mathcal{L}) \subset X(k)$. Then there exists a unique closed reduced subgroup scheme $Z \subset X$ such that $K(\mathcal{L}) = Z(k)$, and $\Lambda(\mathcal{L})|_{X\times Z} = \mathcal{O}_{X\times Z}$.*

Proof. Let $Z = \{x \in X : \Lambda(\mathcal{L})|_{X\times x} = \mathcal{O}\}$, where x not necessarily closed. Then $Z \subset X$ is a closed subset by Q10 (iii), Sheet 3. By 7.3(1), we know $K(\mathcal{L}) = Z(k)$. Give Z the reduced subscheme structure. By Seesaw on $X \times Z$, since $\Lambda(\mathcal{L})|_{e\times X} = \mathcal{O}_X = \Lambda(\mathcal{L})|_{X\times x}$ for all $x \in Z(k)$, $\Lambda(\mathcal{L})|_{X\times Z}$ is trivial. It remains to check Z is a subgroup scheme. As $k = \overline{k}$, and Z is reduced, $Z = \cup Z_i$ is a union of varieties, so $Z \times Z = \cup Z_i \times Z_j$ is reduced, as $Z_i \times Z_j$ is a variety. Consider

$$Z \times Z \hookrightarrow X \times X \xrightarrow{(m,\mathrm{pr}_2)} X \times X$$

, its image is a reduced closed subscheme of $X \times X$, whose k-points are precisely $Z(k) \times Z(k)$. So $Z \times Z \xrightarrow{\sim} Z \times Z \subset X \times X$, $(x, y) \to (x + y, y)$. So Z is a subgroup scheme. □

Remark. *1. In fact, there exists a closed subgroup scheme $\underline{K}(\mathcal{L}) \subset X$ not necessarily reduced, such that for all closed subscheme $S \subset X$, $\Lambda(\mathcal{L})|_{X\times S} \simeq \mathcal{O}_{X\times S} \iff S \subset \underline{K}(\mathcal{L})$, and $Z = \underline{K}(\mathcal{L})^{\mathrm{red}}$*

2. In particular, if $K(\mathcal{L})$ is infinite, then there exists a nonzero abelian subvariety, $Y \subset X$ such that $Y(k) \subset K(\mathcal{L})$. Take Y to be the irreducible component of Z containing the identity.

Let D be an effective divisor on an abelian variety X/k, for $k = \overline{k}$. Namely, $D = \sum_{i=1}^{r} n_i D_i$ for $n_i \geq 0, D_i \subset X$ codimension 1 integral closed subscheme (i.e. subvariety). Note Weil divisor and Cartier divisor agrees as X is smooth over k.

Define $H(D) = \{x \in X(k) \mid T_x^* D = D\}$ be a subgroup of $X(k)$. Note that $T_x^* D = T_{-x} D$. So as $\mathcal{O}_X(T_x^* D) = T_x^* \mathcal{O}_X(D)$, $H(D)$ is a subgroup of $K(\mathcal{O}_X(D)) = \{x \mid T_x^* \mathcal{O}_X(D) = \mathcal{O}_X(D)\}$.

$H(D) = \{k\text{-points of a closed subscheme of } X\}$, altough for more trivial reasons than $K(\mathcal{L})$. In fact, if $Y \subset X$ is a closed subset, and $x \in X(k)$, then:

$$T_x(Y) = Y \iff \forall y \in Y(k), x + y \in Y(k) \iff x \in \bigcap_{y \in Y(k)} \{z \in X \mid (z, y) \in m^{-1}(Y)\}$$

$$\iff x \in \bigcap_{y \in Y(k)} \mathrm{pr}_1(X \times \{y\} \cap m^{-1}(Y))$$

which is closed.

Theorem 7.6 (Ampleness Criterion)**.** *Let D be an effective divisor on X, $\mathcal{L} = \mathcal{O}_X(D)$. The following are equivalent:*

1. *$\mathcal{L}$ is ample*
2. *$K(\mathcal{L})$ is finite*
3. *$H(D)$ is finite*

Corollary 7.7. *An abelian varieties (over any field k) is projective.*

Proof. Let $U \subset X$ any nonempty open affine. Let $D = X \backslash U$ be with reduced subscheme structure. Fact (See Sheet 4): D is a divisor.
Claim: $\mathcal{L} = \mathcal{O}_X(D)$ is ample.
Proof of Claim: By cohomological criterion for ampleness and flat base change, enough to show $\mathcal{L}_{\overline{k}}$ is ample on $X_{\overline{k}}$. Enough to show $H(D)$ is finite. In fact $x \in H(D) \implies T_x(D) = D$, so $T_x(U) = U$ We may assume (translating if necessary) $e \in U(k)$. This implies that $H(D) \subset U(k)$. But if $H(D)$ is infinite, then there exists some $Y \subset X$ closed subscheme of dim > 0, with $Y(k) \subset H(D)$. So $Y(k) \subset U(k) \implies Y \subset U$. But as Y is proper of dimension greater than 0, U is affine, this cannot happen. □

Proposition 7.8. *Let $f : X \to Y$ be a finite morphism. If $\mathcal{L}$ is ample on Y, then $f^*\mathcal{L}$ is ample on X.*

Proof. (of proposition) Since f is finite hence affine, so f_* is exact. Hence, for any coherent $\mathcal{O}_X$-module $\mathcal{F}$, we have:

$$H^p(X, \mathcal{F} \otimes f^*\mathcal{L}^n) = H^p(Y, f_*\mathcal{F} \otimes \mathcal{L}^n) = 0$$

for $i > 0$ an $n \gg 0$. By Serre's Criterion, $f^*\mathcal{L}$ is ample on X. □

Proof. (Of Ampleness Criterion). "2 $\implies$ 3" is clear.

"1 $\implies$ 2": Suppose $\mathcal{L}$ is ample, with $K(\mathcal{L})$ infinite. Then $K(\mathcal{L}) \supset Y(k)$ for a non-zero abelian subvariety of X. As $\mathcal{L}|_Y$ is ample, we may assume $X = Y$. Then $K(\mathcal{L}) = X(k)$. But that means $\mathcal{L} \in \mathrm{Pic}^0(X) = \{\mathcal{L} \mid \varphi_{\mathcal{L}} = 0\}$. But by 7.4(2), $i^*\mathcal{L} = \mathcal{L}^\vee$. So $\mathcal{L}^\vee$ is ample, but this forces $\mathcal{O}_X = \mathcal{L} \otimes \mathcal{L}^\vee$ to be ample. But this means $\dim(X) = 0$, since $H^0(X, \mathcal{O}_X^{\otimes n}) = k$. Contradiction.

"3 $\implies$ 1": Consider $\mathcal{L}^{\otimes 2} = \mathcal{O}_X(2D) \cong T_x^*\mathcal{L} \otimes T_{-x}^*\mathcal{L} = \mathcal{O}_X(T_x^*D + T_{-x}^*D)$, for all $x \in X(k)$ by the Theorem of Square. So for all $x \in X(k)$, there exists $s_x \in H^0(X, \mathcal{L}^{\otimes 2})\backslash\{0\}$ such that $\mathrm{div}(s_x) = T_x^*D + T_{-x}^*D$. Consider $X \to \mathbb{P}^N$ given by $H^0(X, \mathcal{L}^{\otimes 2})$. For any $y \in X(k)$, $y \in T_x^*D + T_{-x}^*D \iff$ one of $y \pm x \in D \iff x \in E_y = T_y^*D \cup i^*T_y^*D$. So for all $x \in (X\backslash E_y)(k)$, $s_x(y) \neq 0$. In particular, for all $y \in X(k)$, there exists $s \in H^0(X, \mathcal{L}^2)$ ssuch that $s(y) \neq 0$. So $\mathcal{L}^2$ gives a morphism $f : X \to \mathbb{P}_k^N$, for $f^*\mathcal{O}_{\mathbb{P}^N}(1) = \mathcal{L}^2$.
Claim: f has finite closed fibres. (i.e. $f^{-1}(p)$ is finite for all $p \in \mathbb{P}^N(k)$)
Proof of Claim: Suppose $f(y) = f(y')$, $y, y' \in X(k)$. Then for all $s \in H^0(X, \mathcal{L}^2)$, either $s(y) = 0 = s(y')$ or $s(y) \neq 0 \neq s(y')$. In particular, if $x \in E_y(k)$, then $s_x(y) = 0 \implies s_x(y') = 0 \implies x \in E_{y'}$. So $E_y = E_{y'}$ as subsets of X. Now if f doesn't have finite fibres, then there exists some irreducible $Y \subset f^{-1}(p)$, for some $p \in \mathbb{P}^N(k)$ with $\dim Y > 0$, and $y, y' \in Y(k)$, $E_y = E_{y'}$. But then for every irreducible component D' of D, $T_y^*D' = T_{y'}^*D'$ (since this holds for $y = y'$, and $T_y^*D \cup i^*T_y^*D = T_y^*D' \cup i^*T_y^*D'$, and Y is connected). So $T_y^*D = T_{y'}^*D$, so $y' - y \in H(D)$ which is infinite. This is a contradiction.

We get $f : X \to \mathbb{P}^N$ is proper (any morphism between proper k-varieties is proper), and has finite fibres. By Zariski's Main Theorem, we know f is finite. We conclude by invoking the previous proposition. □

Corollary 7.9. *Let X/k be an abelian variety, then* $\ker([n] : X(\overline{k}) \to X(\overline{k}))$ *is finite,* $[n] : X \to X$ *is surjective, and the group $X(\overline{k})$ is divisible.*

Proof. We can assume $k = \overline{k}$. Suppose we know $\ker[n]$ is finite. If $x, x' \in X(k)$, then $[n](x) = [n](x') \iff x - x' \in \ker[n]$. So any fibre of the map $[n]$ is finite on each closed point. But if we base change to $k(y)$, we see $[n]$ indeed has finite fibres at all points. So as it's a morphism between varieties of same dimension, it is dominant. As X is proper, $[n]$ is proper, so is closed (and dominant), so $[n]$ is surjective. Then for all $y \in X(k)$, there exists a $x \in X(k)$ with $nx = y$, i.e. $X(k)$ is divisible. Hence it remains to show that the kernel is finite.

Let $\mathcal{L} \in \mathrm{Pic}(X)$ be any ample line bundle. Replacing $\mathcal{L}$ by $\mathcal{L} \otimes i^*\mathcal{L}$, we may assume $\mathcal{L}$ is symmetric. If $\ker[n]$ is infinite, then there exists some closed subvareity V of positive dimension, with $[n](V) = \{e\}$, and the composite

$$V \to X \xrightarrow{[n]} X$$

factors as

$$V \to \mathrm{Spec}\ k \xrightarrow{e} X$$

So the restriction of $[n]^*\mathcal{L}$ to V is trivial. But $[n]^*\mathcal{L} \simeq \mathcal{L}^{\otimes n^2}$ (by 6.6). But $\mathcal{L}$ is ample, this means $\mathcal{O}_V$ is ample $\implies \dim V = 0$. Contradiction. □

This implies that $[n] : X \to X$ is a finite morphism (as in the proof of 7.8 - quasi finite and projective imply finite). As it's a morphism between smooth varieties of the same dimension, it's therefore flat by "Miracle Flatness Theorem". So $[n]_*\mathcal{O}_X$ is therefore a finite flat $\mathcal{O}_X$ module, hence is locally free, whose rank is (by definition) the degree of $[n]$.

Theorem 7.10. $\deg[n] = n^{2g}$, $g = \dim X$. *In particular,* $\#\ker[n](\overline{k}) \leq n^{2g}$,

Proof. Use Hilbert Polynomials. Assume $\mathcal{L} \in \mathrm{Pic}(X)$ symmetric and very ample. So it determines a closed immersion $X \hookrightarrow \mathbb{P}^N$. Consider $\mathcal{F} = [n]_*\mathcal{O}_X$ coherent sheaf on X. We need to compute its rank. We know:

$$P_X([n]_*\mathcal{O}_X, t) = \deg[n] \cdot P_X(\mathcal{O}_X, t) + (\text{lower degrees})$$

But $\deg P_X(\mathcal{O}_X, t) = g = \dim X$. Since $[n] : X \to X$ is finite, so for all $m \in \mathbb{Z}$

$$H^0(X, [n]_*\mathcal{O}_X \otimes \mathcal{L}^m) = H^0(X, [n]^*\mathcal{L}^m) = H^0(X, \mathcal{L}^{mn^2})$$

Hence for $m \gg 0$, we know

$$\begin{aligned}
\deg[n] \cdot P_X(\mathcal{O}_X, m) &= P_X([n]_*\mathcal{O}_X, m) + (\deg < g \text{ in } m) \\
&= \dim H^0(X, [n]_*\mathcal{O}_X \otimes \mathcal{L}^m) + (\deg < g \text{ in } m) \\
&= \dim H^0(X, \mathcal{L}^{mn^2}) + (\deg < g \text{ in } m) \\
&= \dim P_X(mn^2) + (\deg < g \text{ in } m)
\end{aligned}$$

As $\deg P_X = g$, we see $\deg[n] = n^{2g}$. □

It's easy enough to show if n is invertible in $k = \overline{k}$, then $\ker[n] \simeq (\mathbb{Z}/n\mathbb{Z})^{2g}$.

Theorem 7.11. *Assume $k = \overline{k}$, $\mathcal{L}$ is ample. Then $\varphi_{\mathcal{L}} : X(k) \twoheadrightarrow \mathrm{Pic}^0(X)$ is a surjection. So $\mathrm{Pic}^0(X) \simeq X(k)/K(\mathcal{L})$, and carries a structure on an abelian variety of* $\dim = \dim(X)$. *This is called the dual abelian variety.*

Proof. Assume $\dim X > 0$. Let $\mathcal{M} \in \mathrm{Pic}^0(X)$ be not in the image of $\varphi_{\mathcal{L}}$. We compute the cohomology of $\mathcal{F} = \Lambda(\mathcal{L}) \otimes \mathrm{pr}_1^*\mathcal{M}^\vee \in \mathrm{Pic}(X \times X)$ in two different ways.

Let $x \in X(k)$, we see

$$\mathcal{F}|_{X\times x} = T_x^*\mathcal{L} \otimes \mathcal{L}^\vee \otimes \mathcal{M}^\vee = \varphi_{\mathcal{L}}(x) \otimes \mathcal{M}^\vee \in \mathrm{Pic}^0(X)$$

But this is never trivial by assumption. From Sheet 4 Q7, we see $H^p(X, \mathcal{F}|_{X\times x}) = 0$ for all $p \geq 0$. This shows for all open affine $U \subset X$, $H^p(X \times U, \mathcal{F}|_{X\times U}) = 0$. So $H^p(X \times X, \mathcal{F}) = 0$ for all $p \geq 0$, by iterated use of Mayer-Vietoris.

Now slice in the other direction:

$$\mathcal{F}|_{x\times X} = \varphi_{\mathcal{L}}(x) \in \mathrm{Pic}^0(X)$$

If $x \in X(k)\backslash K(\mathcal{L})$, $\mathcal{F}|_{x\times X} \neq \mathcal{O}_X$. So by the same argument, for all $p \geq 0$, $H^p(X, \mathcal{F}|_{x\times X}) = 0$. So for all $U \subset X\backslash K(\mathcal{L})$ and $p \geq 0$, $H^p(U \times X, \mathcal{F}|_{U\times X}) = 0$. But X is projective, so there exists open affine $V \subset X$ with $V \supset K(\mathcal{L})$ [Choose complement of any hyperplane not meeting

the finite set $K(\mathcal{L})$]. By Sheet 2 Q5 again, we know $H^p(V \times X, \mathcal{F}) = H^p(X \times X, \mathcal{F}) = 0$ for all $p \geq 0$.

Then for all $x \in V(k)$, $H^p(X, \mathcal{F}|_{x \times X}) = 0$ for all $p \geq 0$. But $e \in K(\mathcal{L}) \subset V$, and

$$H^0(X, \mathcal{F}|_{e \times X}) = H^0(X, \mathcal{O}_X) = k$$

Contradiction. □